

UNITED STATES PATENT AND TRADEMARK OFFICE

BEFORE THE PATENT TRIAL AND APPEAL BOARD

ALLGENESIS BIOTHERAPEUTICS INC.,
Petitioner,

v.

CLOUDBREAK THERAPEUTICS, LLC,
Patent Owner.

IPR2020-01438
Patent 10,149,820 B2

Before ULRIKE W. JENKS, SHERIDAN K. SNEDDEN, and
ROBERT A. POLLOCK, *Administrative Patent Judges*.

JENKS, *Administrative Patent Judge*.

JUDGMENT

Final Written Decision
Determining No Challenged Claims Unpatentable
35 U.S.C. § 318(a)

I. INTRODUCTION

Allgenesis Biotherapeutics Inc. (“Petitioner”) filed a Petition to institute an *inter partes* review of claims 1–11 (“the challenged claims”) of U.S. Patent No. 10,149,820 B2 (Ex. 1001, “the ’820 patent”), pursuant to 35 U.S.C. §§ 311–319. Paper 1 (“Pet.”). Cloudbreak Therapeutics LLC (“Patent Owner”) filed a Preliminary Response to the Petition. Paper 6.

Considering the record then before us, we determined that Petitioner demonstrated a reasonable likelihood that at least one of the challenged claims was unpatentable. Accordingly, we instituted trial as to claims 1–11 of the ’820 patent. Paper 7 (“Institution Decision” or “Dec.”).

Subsequent to our Institution Decision, Patent Owner filed with the Office a “Disclaimer in Patent Under 37 CFR 1.321(a),” disclaiming claims 1–3 and 6–11 of the ’820 patent. Ex. 2010. Patent Owner also filed a Patent Owner Response. Paper 13 (“PO Resp”). Patent Owner asserted that in view of the statutory disclaimer, only claims 4 and 5 remain in this proceeding. PO Resp. 1 (citing Ex. 2010). Petitioner filed a Reply to Patent Owner’s Response (Paper 22, “Reply”), and Patent Owner filed a Sur-reply (Paper 26, “Sur-reply”).

On December 6, 2021, the parties presented arguments at an oral hearing. The transcript of the hearing has been entered into the record. Paper 28 (“Tr.”).

We have jurisdiction under 35 U.S.C. § 6. This decision is a Final Written Decision, issued pursuant to 35 U.S.C. § 318(a). For the reasons discussed below, we determine that Petitioner has not established by a preponderance of the evidence that claims 4 and 5 of the ’820 patent are unpatentable.

A. Real Parties in Interest

Petitioner identifies itself, “Allgenensis Biotherapeutics Inc.,” as the real party-in-interest. Pet. 13. Patent Owner identifies itself, “Cloudbreak Therapeutics, LLC,” as the real party-in-interest. Paper 3, 2. The parties do not dispute the identification of real parties-in-interest.

B. Related Proceedings

The parties identify U.S. Patent Application No. 16/210,896, filed on December 5, 2018, as a related matter. Pet. 13; Paper 3, 2.

C. Illustrative Claims

In light of Patent Owner’s disclaimer of claims 1–3 and 6–11, only claims 4 and 5 of the ’820 patent remain challenged in the Petition. *See* section I, above; Ex. 2010. But because claims 4 and 5 incorporate all elements of claims from which they depend, we reproduce claims 1, 3, 4, and 5 in their entirety:

1. A method for reducing hyperemia or symptoms thereof in pterygium in an affected eye of a subject in need of such treatment, without surgically excising a pterygium, comprising administering to the affected eye of the subject a therapeutically effective amount of a multikinase inhibitor.
3. The method of claim 1, wherein the multikinase inhibitor is administered to the affected eye in the form of topical ocular formulation or ocular implant.
4. The method of claim 3, wherein the multikinase inhibitor is nintedanib and the nintedanib is administered to the affected eye in the form of a topical ocular formulation and is administered topically to the affected eye.

5. The method of claim 4, wherein the topical ocular formulation is a solution or a suspension.

Ex. 1001, 24:33–37, 48–56.

D. The '820 Patent (Ex. 1001)

1. Specification

The '820 patent is titled “COMPOSITIONS AND METHODS FOR TREATING PTERYGIUM.” Ex. 1001, code (54). The '820 patent describes pterygium as “an ocular surface disease, where an abnormal epithelial and fibroblast growth extends from the nasal or temporal conjunctiva across the limbus onto the cornea. Pterygium patients often experience symptoms of ocular discomfort, hyperemia, and are at risk of visual impairment if the lesion encroaches on the visual axis.” *Id.* at 1:26–31. According to the Specification, “hyperemia” and “ocular hyperemia” “refer[] to redness in an eye resulting from an excess of blood in the white of the eyes (sclera).” *Id.* at 7:19–21. “[R]educing hyperemia,” therefore, indicates a “reduction in redness and/or increase in whiteness in an affected eye.” *Id.* at 7:21–24. The '820 patent further explains that pterygium is “a multifactorial disease” and that “several growth factors such as VEGF and PDGF are potential pathological factors.” *Id.* at 6:49–51

The '820 patent teaches treating early, mid, and late stage pterygium. *See id.* at 10:61–12:30. In one embodiment, the specification “provides a method for treating primary and recurrent pterygium by administering (1) a multikinase inhibitor . . . to the eye of a subject in need of such treatment.” *Id.* at 1:41–45. “[T]he disclosed methods are performed before surgical removal of pterygium, in conjunction with surgery for pterygium removal or

after surgical removal of pterygium, to reduce or prevent pterygium recurrence.” *Id.* at 2:10–14.

The ’820 patent discloses the treatment of human patients (*see id.* at 10:61–11:16), as well as a rabbit cornea suture model (*see id.* at 16:25–19:4 (Example 1)) and a human pterygium mouse model (*see id.* at 19:6–20:45 (Example 2)).

The term “treatment” encompasses any treatment of a disease in a mammal, particularly a human, and includes: (a) preventing the disease and/or symptom(s) from occurring in a subject who may be predisposed to the disease or symptom but has not yet been diagnosed as having it; (b) inhibiting the disease and/or symptom(s), i.e., arresting their development; or (c) relieving the disease symptom(s), i.e., causing regression of the disease and/or symptom(s).

Id. at 8:13–21.

“[T]he term ‘multikinase inhibitor’ (MKI) refers to drug compounds (e.g., a small molecule) that reduce or inhibit the expression or activity of two or more kinases, including, for example, intracellular and/or cell surface protein kinases.” *Id.* at 8:28–32. The multikinase inhibitor is administered to the affected eye in the form of topical ocular formulation, an ointment, a gel, a sustained release semisolid formulation, a sustained release solid formulation or ocular implant. *Id.* at 3:10–15.

Nintedanib and sunitinib are both multikinase inhibitors that inhibit the key VEGFR (vascular endothelial growth factor receptor) and PDGFR (platelet-derived growth factor receptor) families with a few non-overlapping targets. *Id.* at 12:1–2. “Quite unexpectedly, the inventors found that kinase inhibitors with substantial kinase target overlaps can still have different efficacies.” *Id.* at 18:2–5.

Nintedanib showed a clear trend of superiority to sunitinib in reducing neovascular area. While the two kinase inhibitors have substantial target overlaps, distinct efficacies were noted in this rabbit model. Another advantage of nintedanib is that it inhibits a smaller number of targets than sunitinib, allowing for better safety margins and higher doses.

Id. at 11:26–32.

2. Chain of Priority

The '820 patent issued from Application No. 15/991,698 (“the '698 application”), filed May 29, 2018; which is a continuation of Application No. 15/474,620, filed on March 30, 2017, now Patent No. 9,987,223 (“the '223 patent”); which is a continuation of Application No. 15/375,820, filed on December 12, 2016, now Patent No. 9,980,901 (“the '901 patent”); which is a continuation of Application No. PCT/US2016/035726, filed on June 3, 2016; which ultimately claims the benefit of U.S. Provisional Application No. 62/186,660, filed on June 30, 2015 (“the '660 provisional application”); and U.S. Provisional Application No., 62/172,063, filed June 6, 2015 (“the '063 provisional application”). Ex. 1001, codes (60), (63).

Petitioner argues that the '820 patent is not entitled to the benefit date of its earliest provisional application, i.e., the '063 provisional application, because that application fails to provide adequate written description for the breadth of the claims. *See* Pet. 23. We provide an overview of the '063 and '660 provisional applications below.

a) The '063 provisional application (Ex. 1003) having a filing date of June 6, 2015

The '063 provisional application was filed on June 6, 2015. The '063 provisional application explains that “[p]terygium excision with conjunctival

autograft transplantation remains the procedure of choice for definitive treatment of both primary and recurrent pterygia.” Ex. 1003, 8. The abstract of the ’063 provisional application recites:

Methods for reducing hyperemia and symptoms in pterygium patients, stabilizing pterygium, inducing pterygium regression from visual axis/central cornea, and treating pterygium recurrence following pterygiectomy, are disclosed. The methods include administering a multikinase inhibitor (e.g., sunitinib, gefitinib) . . . to a subject.

Ex. 1003, 6, *see also id.* at 12–13 (claims), 15 (Figure 2: Description of different stages (early, middle and late) of pterygium disease development, disease symptoms and suggested treatment at each stage). According to the ’063 provisional application:

This treatment hypothesis is supported by some experimental data conducted in our laboratory, where pterygium vessel regression has been demonstrated with digital imaging following 7 days of three times daily topical ocular dosing of sunitinib, a potent inhibitor of multiple protein kinases of VEGFR2, PDGFR β , KIT, RET, CSFIR and FLT3, in an in vivo rabbit corneal suture model.

Id. at 10. “Similar findings have also been reported in the literature, where topical ocular administration of sunitinib has shown more effective than bevacizumab via a subconjunctival injection, for the inhibition of corneal neovascularization in the rabbit corneal suture model.” *Id.* at 9.

The ’063 provisional application describes that compositions and formulations, including a multikinase inhibitor, can be administered topically or as an implant, or by any other suitable methods known in the art. *Id.* at 11. The ’063 provisional application recognizes that the amount of an agent required for use in treatment will vary with the route of administration

and the nature of the condition for which treatment is required. *Id.*

“Compositions will typically contain an effective amount of the active agent(s), alone or in combination. Preliminary doses can be determined according to animal tests, and the scaling of dosages for human administration can be performed according to art-accepted practices.” *Id.*

b) The '660 provisional application (Ex. 1004) having a filing date of June 30, 2015

The '660 provisional application was filed on June 30, 2015. The '660 provisional application describes that “chronic UV exposure is the single most significant factor in the pathogenesis of pterygium.” Ex. 1004, 9.

Nintedanib is a small molecule that inhibits multiple receptor tyrosine kinases (RTKs) and non-receptor tyrosine kinases (nRTKs). Nintedanib inhibits the following RTKs: platelet-derived growth factor receptor (PDGFR) α and β , fibroblast growth factor receptor (FGFR) 1-3, vascular endothelial growth factor receptor (VEGFR) 1-3, and Fms-like tyrosine kinase-3 (FLT3). Among them, FGFR, PDGFR, and VEGFR have been implicated in pterygium pathogenesis.

Id. at 10. Nintedanib “has shown to be efficacious after three-times daily dosing for 14 days in an in vivo rabbit corneal suture model, an animal model for pterygia, in a prototype ocular solution formulation.” *Id.* at 11. Example 1 describes a 0.1 mg/ml and 0.5 mg/ml nintedanib solution in sterile saline having at a pH 6.0. *Id.* at 13.

E. Asserted Grounds of Unpatentability

Subsequent to the disclaimer of claims 1–3 and 6–11, the following grounds of Petitioner’s alleged unpatentability of claims 4 and 5 remain (Pet. 16):

Claim(s) Challenged	35 U.S.C. §	Reference(s)/Basis
4, 5	102(a)(2)	Allgenesis PCT ¹
4, 5	103	Allgenesis PCT
4, 5	103	King, ² Amparo ³

Petitioner also relies on the Declarations of Dr. Thomas Malone to support its assertions. Ex. 1005; Ex. 1033; *see also* Ex. 2009 (transcript of Dr. Malone’s deposition). Patent Owner relies on the Declaration of Dr. Thomas Hohman in support of its arguments. Ex. 2005; *see also* Ex. 1032 (transcript of Dr. Hohman’s deposition).

II. ANALYSIS

A. *Person of Ordinary Skill in the Art*

Petitioner asserts that a person of ordinary skill in the art would have had at least a doctorate degree in medicinal chemistry, organic chemistry, or related discipline and at least two years of experience in the development of pharmacologic therapies for ophthalmic applications. Pet. 34 (citing Ex. 1005 ¶¶ 34–38). Patent Owner does not contest Petitioner’s definition with respect to the ordinarily skilled artisan. PO Resp. 10. Accordingly, we apply Petitioner’s definition of a person of ordinary skill in the art in your

¹ Nguyen and Lai, WO 2016/209555 A1, published Dec. 29, 2016 (Ex. 1006 (“Allgenesis PCT”)), claiming benefit to U.S. Provisional Application No. 62/183,180 (“the ’180 provisional application”) having a filing date of June 22, 2015 (Ex. 1007).

² King et al., US 2013/0012531 A1, published Jan. 10, 2013 (Ex. 1008).

³ Amparo et al., *Safety and Efficacy of the Multitargeted Receptor Kinase Inhibitor Pazopanib in the Treatment of Corneal Neovascularization*, 54 INVEST. OPHTHALMOL. & VIS. SCI. 537–44 (2013) (Ex. 1009).

analysis. *See, e.g., Okajima v. Bourdeau*, 261 F.3d 1350, 1355 (Fed. Cir. 2001) (“[I]t is always preferable for the factfinder below to specify the level of skill it has found to apply to the invention at issue . . .”).

B. Claim construction

In this *inter partes* review, we construe the claims of the ’820 patent “using the same claim construction standard that would be used to construe the claim in a civil action under 35 U.S.C. § 282(b).” 37 C.F.R. § 42.100(b) (2019). Under that standard, the words of a claim are generally given their “ordinary and customary meaning,” which is the meaning the term would have to a person of ordinary skill at the time of the invention, in the context of the entire patent including the specification. *Phillips v. AWH Corp.*, 415 F.3d 1303, 1312–13 (Fed. Cir. 2005) (en banc).

Petitioner offers proposed claim constructions for the preamble and the phrase “therapeutically effective amount.” *See* Pet. 35–39.

1. Preamble

The preamble of claim 1 recites: “A method for reducing hyperemia or symptoms thereof in pterygium in an affected eye of a subject in need of such treatment, without surgically excising a pterygium.” Ex. 1001, 24:33–35.

Petitioner offers a proposed claim construction that suggests only the first portion of the preamble is limiting, and the negative limitation “without surgically excising a pterygium” is not limiting because it is a statement of intended use. Pet. 35. Petitioner argues that just because some preamble language is limiting does not mean that the entire preamble is limiting. Pet. 38 (citing *Cochlear Bone Anchored Sols. AB v. Oticon Med. AB*, 958

F.3d 1348, 1355 (Fed. Cir. 2020) (citing *TomTom, Inc. v. Adolph*, 790 F.3d 1315, 1323 (Fed. Cir. 2015))). Petitioner therefore argues that the clause “without surgically excising a pterygium” should be accorded no weight, because the clause does not provide an antecedent basis and is unnecessary to give life or meaning to the claim. The ’820 patent states that “the disclosed methods are performed before surgical removal of pterygium, in conjunction with surgery for pterygium removal or after surgical removal of pterygium, to reduce or prevent pterygium recurrence.” Ex. 1001, 2:10–14. Petitioner’s position is that the methods can be performed regardless of surgical intervention and, therefore, the clause does not further limit the claim. Pet. 38.

In our Institution Decision, we found that the negative limitation in the preamble “without surgically excising a pterygium” is limiting because it describes the patient population that is receiving the treatment. Dec. 15. The preamble therefore is necessary to give life and meaning to the method of applying the composition to a particularly defined patient population. The ’820 patent specification provides that the treatment can be applied to patients before, during, or after surgery. Ex. 1001, 2:10–13, 12:24–27; *see also* Ex. 1003, 10 (“At the mid stage of pterygium, where fibrovascular growth extends from conjunctiva to limbus, the goal is to stabilize pterygium progression to delay or to avoid surgery for removing pterygium or even to induce pterygium regression from visual axis / central cornea.”). The specification, therefore, describes a larger patient group than the patient group recited in the claims. Neither Petitioner nor Patent Owner has responded to our preliminary construction of the negative limitation recited in the preamble, *see generally* Reply; PO Resp. 8–9, nor do we see any

reason to modify it here. Accordingly, we maintain our construction of the preamble for this Decision. Specifically, we find that the negative limitation “without surgically excising a pterygium” is not an intended use statement but instead defines a particular patient population receiving the treatment. Dec. 15.

2. *“Therapeutically effective amount”*

Petitioner contends that “therapeutically effective amount” refers to an amount of an agent effective to produce the intended result. Pet. 39.

According to the ’820 patent, “the terms ‘therapeutically effective’ and ‘effective amount’ refer to that amount of an agent effective to produce the intended pharmacological, therapeutic or preventive result.” Ex. 1001, 7:35–38. The ’820 patent further explains that “[t]herapeutically effective dosages in the methods described herein can be determined by the treating physician.” *Id.* at 7:50–52.

In our Institution Decision, we found “therapeutically effective” and “effective amount” are reasonably construed as the amount of an agent that will produce the intended result as assessed by the person of ordinary skill in the art. Dec. 16. Petitioner does not address our construction in the Reply. *See generally* Reply. Patent Owner also does not contest this construction. PO Resp. 10. Accordingly, we maintain our construction for this Decision.

C. *Overview of the asserted prior art*

1. *Overview of Allgenesis PCT (Ex. 1006)*

Allgenesis PCT was filed May 27, 2016, and has a publication date of December 29, 2016. Ex. 1006, code (22), (43). Allgenesis PCT claims

priority to the '180 provisional application, filed June 22, 2015. *Id.* at code (30); *see also* Ex. 1007, 1.

Allgenesis PCT is directed to ophthalmic formulations containing tyrosine kinase inhibitors, such as Nintedanib, Axitinib, Sorafenib, and Pazopanib for treating conditions such as pterygium. Ex. 1006, 1 (code (57)). A “tyrosine kinase inhibitor (TKi)” is described as “a therapeutically active compound that is capable of inhibiting the activity of one or more tyrosine kinases, such as, for example, platelet-derived growth factor receptor (PDGFR), fibroblast growth factor receptor (FGFR), vascular endothelial growth factor receptor (VEGFR), and Fms-like tyrosine kinase-3 (FLT3).” *Id.* at 7.

“Pterygium is a fleshy lesion or growth originating from the conjunctiva of the eye.” *Id.* at 2. “[T]he ocular surface disease to be treated [using the tyrosine kinase inhibitor] is hyperemia associated with pterygium, pterygium conjunctivae, recurrent pterygium, or corneal angiogenesis.” *Id.* at 17. “Hyperemia (redness) generally refers to an increase in blood flow to the eye resulting in increased redness of the eye. Hyperemia can occur on its own (primary), or it can be a symptom associated with one or more other ocular diseases (secondary), such as pterygium.” *Id.*

One embodiment of Allgenesis PCT is an ophthalmic formulation comprising:

- (a) a therapeutically effective amount of nanoparticles comprising Nintedanib or a pharmaceutically acceptable salt thereof in a concentration of about 0.1 % w/v to about 10% w/v;
- (b) about 0.01% to 0.3% tyloxapol; and
- (c) about 0.1 % to 1% HPMC

wherein the formulation is a liquid suspension for topical administration.

Id. at 21; *see also id.* at 22 (example 1), 27 (example 3). Allgenesis PCT teaches the application of an ophthalmic suspension to the eye of an animal, using a cornea suture-induced neovascularization (CNV) model. *Id.* at 29–31. The results of the “study demonstrate that ophthalmic suspension formulations containing Nintedanib Ethanesulfonate and Axitinib were both well-tolerated and efficacious in preventing or slowing CNV progression. The results also demonstrate that Nintedanib Ethanesulfonate was more efficacious, or at least comparable to Avastin in preventing or slowing CNV progression.” *Id.* at 31, *see also id.* at 36–37 (claims 12–15).

2. Overview of King (Ex. 1008)

King is a U.S. Patent Application Publication US 2013/0012531 A1, published on January 10, 2013, from U.S. Patent Application No. 13/518,407, filed on January 5, 2011. Ex. 1008, codes (21), (22), (43). King describes methods of treating disorders of ocular angiogenesis or vascular leakage in a patient by administration of inhibitors such as pazopanib. *Id.* at code (57). “[T]he disorder of ocular angiogenesis or vascular leakage can be edema or neovascularization for any occlusive or inflammatory retinal vascular disease, . . . [including] pterygium.” *Id.* ¶ 28. King discloses that “‘treatment’ means any manner in which one or more symptoms associated with the disorder are beneficially altered.” *Id.* ¶ 15. King teaches that methods of treating disorders of ocular angiogenesis include “orally administering to the patient between 1 and 50 mg of a suitable inhibitor.” *Id.* ¶¶ 4, 18.

King discloses a list of inhibitors that target one or more of the following receptors VEGFR1, VEGFR2, VEGFR3, PDGFRalpha, PDGFRbeta, c-kit, and/or FGFR. The list of King's inhibitors includes pazopanib (*id.* ¶¶ 19, 22) as well as the following multikinase inhibitors:

apatinib, sunitinib, sorafenib, bivanib, midostaurin (PKC412) (an inhibitor of FLT3, c-KIT, VEGFR-2, PDGFR and multiple isoforms of the serine/threonine protein kinase C (Pkc), under development by Novartis), E-7050 [golvatinib] (a C-met and VEGFR tyrosine kinase inhibitor, under development by Eisai), XL-184 a spectrum selective kinase inhibitor that inhibits Met, Ret and VEGFR2, under development by Exelixis), XL-647 [tesevatinib] (an orally-available tyrosine kinase inhibitor under development by Exelixis that inhibits EGFR, HER2, VEGFR and EphB4), cediranib, linifanib, motesanib, RAF-265 (formerly CHIR-265) a B-Raf and VEGFR kinase inhibitor, under development by Novartis), tivozanib, TAK-593 (a VEGFR/PDGFR tyrosine kinase inhibitor, under development by Millennium (Takeda)), ARQ-197 [tivantinib] (an ATP-independent inhibitor of c-Met under development by ArQule (Cyclis Pharmaceuticals before the acquisition)), OSI-930 (a c-kit and VEGFR-2 tyrosine kinase inhibitor under development by OSI Pharmaceuticals), DCC-2036 [rebastinib] (a Bcr-abl inhibitor inhibits that also inhibits the Src-like kinases LYN, HCK and FGR, as well as the TIE2 and KDR kinases, under development by Deciphera Pharmaceuticals), MGCD-265 (an inhibitor of c-Met, VEGFR1, VEGFR2, VEGFR3, Tie-2 and Ron tyrosine receptor kinases, under development by MethylGene, in collaboration with ChemBridge Research Laboratories, CA, the US), PF-337210 (an inhibitor of VEGFR2, under development of Pfizer), *BIBF-1120^[4] [nintedanib] (a VEGFR-2, PDGF and FGF kinase inhibitor, which also inhibits the src, lck and lyn tyrosine kinases, under development by Boehringer Ingelheim)*, ENMD-2076 (a kinase inhibitor that

⁴ Nintedanib (also known as BIBF-1120) is a multikinase inhibitor developed by Boehringer Ingelheim. Pet. 19 (citing Ex. 1011, 1053).

selectively targets aurora kinase A vs B, and also inhibits Flt3, c-kit, CSF1R and KDR (VEGFR2) as well as VEGFR3, PDGFR-alpha, FGFR1, FGFR2, EphA1 and src, under development by EntreMed), TG-100-801 (a VEGFR, Src, Yes, Lek, Lyn kinases and PDGFR- β inhibitor, under development by TargeGen), BMS-690514 (an inhibitor of EGFR, HER2, ErbB4 and VEGFR1-3, under development by Bristol-Myers Squibb), SSR-106462 (a TIE-2 inhibitor and VEGFR-2 tyrosine kinase inhibitor, under development by SanofiAventis), BAY-73-4506 [regorafenib](a VEGFR, KIT, RET, FGFR and PDGFR kinase inhibitor, under development by Bayer), plitidepsin, axitinib, vandetinib, and nilotinib.

Ex. 1008 ¶ 19 (emphasis added).

King's experimental procedure includes topically applying pazopanib inhibitors in a rat animal model. *Id.* ¶ 50. "This methodology has been commonly applied in experimental studies of neovascular [age-related macular degeneration] AMD and allows predictions to be made on drug efficacy in humans. In particular, VEGF expression becomes upregulated and effects of VEGFR as well as PDGFR tyrosine kinase receptors can be well predicted since such like antagonists inhibit" choroidal neovascularization. *Id.* ¶ 58.

King discloses using "[t]opical eye drops . . . formulated in a buffered 7% cyclodextrin solution containing 5 mg/ml pazopanib." *Id.* ¶ 48. "Pazopanib (approximately 30 μ l of a 5 mg/ml sterile filtered solution) was administered twice daily [to the affected eye of the animals]. Animals of the control group received a vehicle only." *Id.* ¶ 50. "[T]opically administered pazopanib significantly reduced development" of choroidal neovascularization lesions. *Id.* ¶ 59. "These results indicated that a twice-

daily topical administration of pazopanib inhibited further lesion development by >89%.” *Id.* ¶ 60.

3. *Overview of Amparo (Ex. 1009)*

Amparo is a publication titled “Safety and Efficacy of the Multitargeted Receptor Kinase Inhibitor Pazopanib in the Treatment of Corneal Neovascularization” having a publication date of January 2013. Ex. 1009, 1.

Amparo discloses that “VEGF and PDGF play critical roles in corneal neovascularization. . . . Blockade of VEGF signaling suppresses vascular endothelial cell migration, proliferation, and survival. Pazopanib, a multitargeted receptor kinase inhibitor, blocks phosphorylation signaling induced by ligand binding to these receptors.” *Id.* at 6 (citations omitted).

The purpose of Amparo’s study was “[t]o evaluate the safety and efficacy of topical pazopanib in the treatment of corneal neovascularization (CNV).” *Id.* at 1. “Only patients with clinically stable CNV that extended at least 1-mm beyond the limbus were considered for enrollment.” *Id.* at 2. Patients were instructed to apply one drop of pazopanib 0.5% solution into the eye four times per day for three weeks. *Id.* The patient population tested included a patient having a diagnosis of “MGD, rosacea, DES, [and] recurrent pterygium.” *Id.* at 3 (table 2, patient 15). “The primary efficacy outcome measures were the size and extent of CNV. The efficacy of topical pazopanib treatment on CNV was evaluated by comparing corneal photographs acquired at baseline with photographs acquired on the follow-up visits.” *Id.* at 2. “Treatment with topical pazopanib led to a 22% decrease of corneal neovessels (NA) and NA is a function of vessel caliber and

length.” *Id.* at 6. The study shows that “topical pazopanib is safe and well tolerated, and can effectively reduce established CNV as measured by invasion area, vessel length, and neovascular area.” *Id.*

D. Alleged Anticipation over Allgenesis PCT

Petitioner provides a detailed explanation to show that Allgenesis PCT sets forth each limitation of challenged claims 4 and 5 to a person of ordinary skill in the art. Pet. 49–58. Patent Owner opposes. PO Resp. 11–23.

According to Petitioner, Allgenesis PCT is subject to the first inventor to file provisions of the AIA. *See* Pet. 40 n.3. Thus, Petitioner contends that, for the purpose of applying Allgenesis PCT⁵ as art, the earliest effective filing date of Allgenesis PCT is June 22, 2015. Patent Owner does not appear to dispute Allgenesis PCT effective filing date. *See* PO Reply 1, 11.

Whether Allgenesis PCT is prior art to the ’820 patent depends on whether the ’820 patent is entitled to its earliest effective filing date of June 6, 2015. Petitioner contends that the ’820 patent is not entitled to the June 6, 2015 filing date of the ’063 provisional application, because the ’063 provisional application does not provide sufficient written descriptive support or enablement under 35 U.S.C. § 112 for the claims of the ’820 patent. *See* Pet. 23–34. Additionally, Petitioner contends that the Office previously determined that “Patent Owner’s June 6, 2015 priority document does not provide sufficient written description or enablement for the subject

⁵ Petitioner provides a claim chart showing where each limitation of claim 1 of Allgenesis PCT finds support in the ’180 provisional application, which has a filing date of June 22, 2015. *See* Pet. 41–42.

matter in its later-published nonprovisional applications.” Pet. 33 (citing Ex. 1015, 619–27, 637–54, 678–82).

1. The earliest priority date of the ’820 patent

A claim for priority under 35 U.S.C. § 119(e) requires the patent application claiming priority be “disclosed in the manner provided by section 112(a) (other than the requirement to disclose the best mode) in [the] provisional application” to which priority is claimed. For the ’820 patent to have an effective filing date of June 6, 2015, the ’063 provisional application must describe (providing written description and enablement support for) the subject matter of the disputed claims. MPEP § 2151 (9th ed., rev. 10.2019, June 2020) (“a U.S. patent document is effective as prior art as of the filing date of the earliest application to which benefit or priority is claimed and which describes the subject matter relied upon, regardless of whether the earliest such application is a U.S. provisional or non-provisional application”).

The Federal Circuit has explained the conditions under which an issued patent is entitled to the filing date of a provisional application for prior art purposes, stating

for a non-provisional application to claim priority to a provisional application for prior art purposes, ‘the specification of the provisional [application] must contain a written description of the invention . . . in such full, clear, concise, and exact terms, to enable an ordinarily skilled artisan to practice the invention claimed in the non-provisional application.’ Further, we have previously stated that ‘for the non-provisional utility application to be afforded the priority date of the provisional application . . . the written description of the provisional must adequately support the claims of the non-provisional application.

Amgen Inc. v. Sanofi, 872 F.3d 1367, 1380 (Fed. Cir. 2017) (quoting *Dynamic Drinkware, LLC v. National Graphics, Inc.*, 800 F.3d 1375, 1378 (Fed. Cir. 2015) and *New Railhead Mfg., LLC v. Vermeer Mfg. Co.*, 298 F.3d 1290, 1294 (Fed. Cir. 2002)) (alterations in original).

a) The state of the art at the time of filing the '063 provisional application

Before we address the parties' arguments with respect to the written description and enablement issues raised in the Petition, we find it helpful to review the state of the art before the filing date of the '063 provisional application.

(i) King (Ex. 1008) discloses using the multikinase inhibitor pazopanib as an eye drop

King identifies twenty-seven multikinase inhibitors as suitable inhibitors for treating disorders of ocular angiogenesis. *See above* II.C.2. The list of inhibitors includes sunitinib, nintedanib (BIBF-1120), and pazopanib. Ex. 1008 ¶¶ 19–20, code (57). King formulates and tests one multikinase inhibitor – pazopanib – as an eye drop. Pazopanib is formulated at a 5 mg/ml concentration in a buffered 7% cyclodextrin solution. *Id.* ¶ 48. King administered 30 µl of this 5 mg/ml pazopanib solution onto the eyes of test animals. *Id.* ¶ 50. King concludes that topically administered pazopanib significantly reduced development of choroidal neovascularization lesions. *See id.* ¶¶ 58–62.

(ii) Amparo (Ex. 1009) discloses using the multikinase inhibitor pazopanib as an eye drop

Amparo describes that “[p]azopanib (GlaxoSmithKline, King of Prussia, PA), is a small multi-tyrosine kinase inhibitor (TKI) that targets

VEGF and PDGF receptors and inhibits angiogenesis.” Ex. 1009, 537; *see also above* II.C.3. Amparo administers a 0.5% pazopanib⁶ solution (GlaxoSmithKline) onto the eyes of test subjects. *Id.* at 538. Amparo concludes that the results of this study indicate that topical treatment with pazopanib 0.5% leads to a statistically significant reduction (22%) in corneal neovascularization (CNV) as assed by measuring the neovascular area (NA), vessel length (VL), and invasion area (IA). *Id.* at 542.

(iii) Peyman⁷ (Ex. 1010) discloses using the multikinase sunitinib as an eye drop

Peyman delivers anti-vascular endothelial growth factor (VEGF) agents such bevacizumab, ranibizumab, pegaptanib, sunitinib maleate, TNP470, integrin av antagonists, 2-methoxyestradiol, paclitaxel, or anti-VEGF siRNA under conditions sufficient to ameliorate a cause and/or effect of fluid leakage in the eye. Ex. 1010 ¶ 1. Peyman exemplifies using bevacizumab (a monoclonal antibody that inhibits the biologic activity of human VEGF) and sunitinib maleate (a small molecule multikinase inhibitor) in an animal model for corneal neovascularization (CNV). *See id.* ¶¶ 11, 121–147. Peyman concludes that both sunitinib maleate and bevacizumab reduces corneal neovascularization in the treated eyes. *Id.* ¶¶ 142–147.

⁶ This is equivalent to a 5 mg/ml pazopanib solution.

⁷ Peyman, WO 2007/038453 A2, published April 5, 2007 (Ex. 1010).

- (iv) Roth⁸ (Ex. 1011) discloses that nintedanib functions similarly to sunitinib and sorafenib in systemic therapy applications

Roth discloses that “nintedanib (BIBF1120) is a potent, oral, small-molecule tyrosine kinase inhibitor, also known as a triple angiokinase inhibitor, inhibiting three major signaling pathways involved in angiogenesis.” Ex. 1011, Abstract. Roth discloses nintedanib’s various kinase targets. *Id.* at 1055 (Table 2).

The clinical success of bevacizumab has established the principle of antiangiogenesis as a valid treatment strategy in oncology. . . . Antiangiogenic tyrosine kinase inhibitors have also been integrated into standard of care clinical practice for patients with other cancers, such as sunitinib, sorafenib, pazopanib or axitinib in [human renal cell carcinoma] RCC, and sorafenib in hepatocellular carcinoma (HCC) and differentiated thyroid carcinoma.

Id. at 1056. “Nintedanib 200 mg b.i.d. monotherapy demonstrated comparable efficacy versus standard of care sunitinib as first line systemic therapy in a randomized (2:1) phase II study in patients with advanced [human renal cell carcinoma] RCC.” *Id.* at 1055. Similar results were obtained when nintedanib was compared to sorafenib. *Id.*

- (v) Pérez-Santonja⁹ (Ex. 1014) discloses using the multikinase inhibitor sunitinib as an eye drop

Pérez-Santonja discloses inhibiting corneal neovascularization by topically applying bevacizumab and sunitinib. Ex. 1014, Abstract. Pérez-

⁸ Roth et al., *Nintedanib: From Discovery to the Clinic*, 58 J. Med. Chem. 1053–1063 (2015), published December 4, 2014 (Ex. 1011).

⁹ Pérez-Santonja et al., *Inhibition of Corneal Neovascularization by Topical Bevacizumab (Anti-VEGF) and Sunitinib (Anti-VEGF and anti-PDGF) in an Animal Model*, 150 Am J Ophthalmol. 519–528 (2010) (Ex. 1014).

Santonja uses bevacizumab at a concentration of 5 mg/ml and sunitinib at a concentration of 0.5 mg/ml. *Id.* at 520. “Topical sunitinib consisted of sunitinib malate powder (99% chemical purity) mixed with sterile saline.” *Id.* at 521. “[I]n comparison with saline, corneal neovascularization was inhibited 27.2% by bevacizumab on day 14 and 81.9% by sunitinib.” *Id.* at 524.

(vi) Davis (Ex. 1016) discloses a list of 72 multikinase inhibitors

Davis discloses 72 multikinase inhibitors, twenty-four of which overlap with the multikinase inhibitors listed in the ’820 patent. *Compare* Ex. 1001, 8:47–56 and Ex. 1003, 12 (claims 1–3) with Ex. 1016, 17–27 (Supplemental Table 2) and Ex. 1008 ¶ 19; Ex. 1005 ¶ 66. Of the forty-eight inhibitors listed in Davis but not included in the list of inhibitors of the ’820 patent, six inhibitors are identified as having activity against VEGFR2. Ex. 1016, 17–27 (Supplementary table 2). The six inhibitors are: brivanib (BMS-540215), linifanib (ABT-869), midestaurin (PKC-412), motesanib (AMG-706), Ki-20227, and SU-14813. *See id.*

(vii) Summary

A review of prior art shows that the list of multikinase inhibitors presented in the ’063 provisional application and recited in the ’820 patent comprise a selected subset of inhibitors, and not simply a list of all known multikinase inhibitors at the time of filing the ’063 provisional application. *Compare* Ex. 1001, 8:47–56 and Ex. 1003, 12 (claims 1–3) with Ex. 1016, 17–27 (Supplemental Table 2) and Ex. 1008 ¶ 19. This is supported by Petitioner’s expert, Dr. Malone, who acknowledges that the ’820 patent has twenty-four multikinase inhibitors that overlap with the 72 multikinase

inhibitors disclosed in Davis and notes that there are additional multikinase inhibitors that are included in the '820 patent that are not studied by Davis. Ex. 1005 ¶ 66, *see also id.* n.1 (“Compounds identified in the '820 Patent and the June 6, 2015 Provisional application but not studied in Davis et al. [Ex. 1016] are Amuvatinib, Cabozantinib, Ceritinib, Crenolanib, Dabrafenib, Dacomitinib, Golvatinib, Ibrutinib, Icotinib, Idelalisib, Lenvatinib, Palbociclib, Ponatinib, Regorafenib*, Tivantinib, Tivozanib*, Trametinib, and Vemurafenib.”).

The list of multikinase inhibitors presented in the '820 patent, and disclosed in the '063 provisional application, is also not a list of all known compounds that have activity against a combination of VEGFR (1, 2, 3), PDGFR (α , β), and FGFR (1, 2, 3) recited in the '820 patent as useful for the disclosed methods. *See* Ex. 1001, 8:40–46. Davis discloses that BMS-540215 (brivanib) is a compound that has primary activity against VEGFR2, PDGFR A/B, as well as activity against FGFR1 and FGFR2. *See* Ex. 1016, 30 (Supplemental Table 2), 40 (Supplemental Table 3). Thus, brivanib (BMS-540215) has activity against the combination of targets identified in the '820 patent but is not included in the list of multikinase inhibitors in the '820 patent or '063 provisional application. *See* Ex. 1001, 8:47–56; Ex. 1003, 12 (claim 3).

The list of multikinase inhibitors presented in the '820 patent, and disclosed in the '063 provisional application, is also not a list of all multikinase inhibitors known at the time of filing the provisional application to target VEGFR2. Specifically, BMS-540215 (brivanib), linifanib, midostaurin (PKC-412), motesanib (AMG-706), KI-20227, SU-14813, OSI-930, MGCD-265, PF-337210, ENMD-2076, and BMS-690514 are disclosed

as having activity against VEGFR2 but are not included in the list recited in the '820 patent or the '063 provisional application. *See* Ex. 1016, 17–27 ((Supplemental table 2) (listing BMS-540215 (brivanib), linifanib, midostaurin, motesanib, Ki-20227, SU-14813)); Ex. 1008 ¶ 19 (listing OSI-930, MGCD-265, PF-337210, ENMD-2076, BMS-690514, brivanib, linifanib, midostaurin, motesanib).

Based on these disclosures in the art, we determine that the list of inhibitors in the '820 patent and '063 provisional application is not simply a list of *all known* multikinase inhibitors at the time of filing the provisional application, nor is it a list of all known inhibitors that target a particular receptor such as VEGFR2. The list of inhibitors included in the '820 patent, and recited '063 provisional application, therefore, is a subset of inhibitors known at the time of filing the invention. *See* Tr. 47:16–48:7.

We further note that at the time of filing the '063 provisional application, it was known how to test multikinase inhibitors for their ability to inhibit vascularization. *See* Ex. 1008; Ex. 1009; Ex. 1010; Ex. 1014. For example, King tested topical application of pazopanib eye drops using a choroidal neovascularization assay in rats. *See* Ex. 1008 ¶¶ 58–62. Amparo exemplified testing the multikinase inhibitor pazopanib in the form of eye drops in human subjects. Ex. 1009. Moreover, Peyman and Pérez-Santonja exemplify testing the multikinase inhibitor sunitinib as eye drops for treating corneal neovascularization. *See* Ex. 1010; Ex. 1014. We note that out of all the references, only Pérez-Santonja details how the multikinase inhibitor sunitinib is formulated as an eye drop. Specifically, “[t]opical sunitinib [of Pérez-Santonja] consisted of sunitinib malate powder (99% chemical purity) mixed with sterile saline.” Ex. 1014, 521.

The above review presents a baseline for evaluating the arguments presented by the parties.

b) Written description

After instituting trial,¹⁰ Patent Owner filed a disclaimer disclaiming the generic subject matter, and the remaining claims 4 and 5 are narrowly tailored to a single multikinase inhibitor – nintedanib. *See* Ex. 1001, 24:51–56 (claims 4 and 5); PO Resp. 1; Ex. 2010.

The issue before us is whether the '063 provisional application provides sufficient written descriptive support for a method of reducing hyperemia or symptoms thereof in pterygium in an affected eye of a subject by applying a therapeutically effective amount of the multikinase inhibitor nintedanib to the affected eye of the subject, as recited in claims 4 and 5.

¹⁰ Most of the claims at issue in the Institution Decision were very broad, because they were directed to “multikinase inhibitors” generally that encompassed a large genus of inhibitors including inhibitors that have not yet been described or discovered. *See* Dec. 26–27. Our conclusion in the Institution Decision was that the '063 provisional application does not have written descriptive support for the broad claims to the genus because it did not provide a means to distinguish which inhibitors are encompassed by the genus. In *Juno*, our reviewing court explained that a sufficient written description for a genus of compounds, “which [in that case] encompasses all scFvs, known and unknown, as part of the claimed CAR that bind to a selected target,” requires that the specification must provide means for distinguishing which scFvs bind to which targets. *See Juno Therapeutics, Inc. v. Kite Pharma, Inc.*, 10 F.4th 1330, 1338 (Fed. Cir. 2021). In *Juno*, “only four or five CD19 scFv species [were] known at the time,” and because the specification did not provide a way to distinguish which scFvs belong to the group that binds the target, there was a lack of written descriptive support for the genus. *Id.* at 1341. The facts in *Juno* differ from the present proceeding because the remaining claims are now narrowly tailored to a single multikinase inhibitor – nintedanib.

(i) Petitioner's Contentions

Petitioner contends that “[t]he June 6, 2015 provisional contains reference to testing for only one multikinase inhibitor (sunitinib). This one experiment—which is described with little detail and for which no data is actually shown—reflects a presumed reduction to practice of only a single species of multikinase inhibitor.” Pet. 26. Petitioner specifically focuses on the “treatment hypothesis” or “hypothesized several strategies” language used in the ’063 provisional application as evidencing that the inventors did not possess the claimed invention as of the June 6, 2015, filing date. Pet. 26–27 (citing *Ariad Pharm., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1352–53 (Fed. Cir. 2010) (“[A] description that merely renders the invention obvious does not satisfy the [written description] requirement.”)); Reply 12 (“The ’063 provisional discloses no testing of nintedanib or any of the other 38 listed multikinase inhibitors.”). Petitioner contends that “[i]t was not until Patent Owner’s later provisional application filed on June 30, 2015—after the effective date for the Allgenensis PCT—that Patent Owner first disclosed testing showing efficacy for nintedanib.” Reply 13.

Petitioner contends that Patent Owner is on the record in the prosecution of a continuation application of the ’820 patent as admitting that multikinase inhibitors “are not equivalent or substitutable.” Pet. 27–28 (citing Ex. 1023,¹¹ 9 (“all anti-VEGF agents are not equivalent or substitutable.”). Petitioner’s expert, Dr. Malone, avers that a person of ordinary skill in the art “would not have considered efficacy of one

¹¹ This is a Request for Continued Examination and Reply to Office Action in continuing Application No. 16/210,896, filed April 10, 2020, which claims benefit to the ’820 patent.

multikinase inhibitor to be representative of the activity of others.” Pet. 28 (citing Ex. 1005 ¶¶ 62–71). Petitioner contends that a person of ordinary skill in the art would not assume that “inhibitors with different structures, different targets, and different mechanisms of action” would behave similarly. Pet. 29 (citing Ex. 1005 ¶¶ 62–72; Ex. 1022, 2–3, 10).

Petitioner argues that the ’063 provisional application has not exemplified the use of nintedanib; therefore, the ’063 provisional application cannot provide written descriptive support for claims 4 and 5 of the ’820 patent that are directed to methods of treating pterygium using nintedanib. *See* Pet. 25–30; Reply 11–13 (citing *Nuvo Pharms. (Ireland) Designated Activity Co. v. Dr. Reddy’s Labs. Inc.*, 923 F.3d 1368, 1380 (Fed. Cir. 2019)); Ex. 1005. Specifically, Petitioner contends that “[t]he inventor was merely guessing that *any* of his proposed treatment methods would have efficacy treating pterygium, let alone a method using nintedanib specifically.” Reply 12–13. Petitioner’s expert, Dr. Malone, avers that the ’063 provisional application does not describe testing data or any other functional similarities between sunitinib and nintedanib. Ex. 1005 ¶ 62. “The vast majority of small-molecule kinase inhibitors interact with multiple members of the protein kinase family and thus are MKIs.” *Id.* ¶ 65 (citing Ex. 1016, 1046). According to Dr. Malone, the physical structures and properties of just these twenty-four inhibitors identified in the ’820 patent vary greatly, and “I do not think that a person skilled in formulating drugs for ocular delivery could consider these compounds to be equivalent in the absence of actually formulating each drug and testing for efficacy.” Ex. 1005 ¶ 68, *see id.* ¶¶ 70–71; Ex. 1033 ¶ 27. Dr. Malone contends that

“[s]ubstantial effort was required [by Pérez-Santonja] to perform one test using a single multikinase inhibitor [sunitinib].” Ex. 1005 ¶ 74.

Dr. Malone avers that “[t]he pathogenesis of pterygium is complex, and many factors have been identified that play a role in its progression. . . . Angiogenic factors play an important role in the development of pterygium, particularly in the development of blood vessels needed to supply nutrients to the lesion and that contribute to hyperemia experienced by patients with the condition.” Ex. 1033 ¶ 19; *see also id.* ¶ 22 (“It is my opinion that many growth factors and receptors contribute to angiogenesis in pterygium, not just VEGFR2.”). Dr. Malone disagrees with Patent Owner’s expert, Dr. Hohman. Specifically, Dr. Malone opines that “[s]unitinib and nintedanib have different and unique inhibition profiles, which would have been readily apparent and known to a person of ordinary skill in the art.” *Id.* ¶ 23. Relying on the inhibition profiles recited in the ’820 patent, Dr. Malone avers that “[s]unitinib inhibits several kinase targets that are not identified as targets of nintedanib, including c-Kit, c-Fms, and RET. Nintedanib inhibits several kinase targets that are not identified as targets of sunitinib, including FGFR4, Lyn, and Src. And for the overlapping kinase targets, sunitinib and nintedanib have different inhibition potencies,” and a person of ordinary skill in the art would have understood the inhibitors are not equivalent. *Id.* ¶ 25.

(ii) Patent Owner’s Contentions

Patent Owner contends that the claims “are limited to applying topical formulations of a specific multikinase inhibitor, nintedanib, to treat hyperemia associated with pterygia.” PO Resp. 13. According to Patent Owner, the ’063 provisional application discloses the use of nintedanib to

treat hyperemia associated with pterygia. At the early stage of pterygium, during which patients experience “ocular discomfort, hyperemia, irritation, blurred vision, foreign body sensation or pain, the treatment goal is to reduce hyperemia and symptoms. To achieve this goal, we have invented a treatment of using a multikinase inhibitor in a proper ocular dosage form.”

PO Resp. 14 (citing Ex. 1003, 9).

This treatment hypothesis is supported by some experimental data conducted in our laboratory, where sunitinib, a potent inhibitor of multiple protein kinases of VEGFR2, PDGFRP, KIT, RET, CSF1R and FLT3, has been shown to be efficacious after three-times daily dosing for 14 days in an in vivo rabbit corneal suture model, an animal model for pterygia, in a prototype ocular solution formulation.

Ex. 1003, 10; PO Resp. 15. Patent Owner’s expert, “Dr. Hohman[,], explains[that] a person of skill would understand from this that because sunitinib worked in an animal model for pterygia, so too would nintedanib because both are multikinase inhibitors with very similar pharmacological potencies related to the VEGF receptors.” PO Resp. 15 (citing Ex. 2005 ¶ 46). Patent Owner contends that “[t]he key target receptor for pterygium hyperemia is VEGFR2 and the inhibition potency, expressed as IC₅₀, is 10 nM for sunitinib, and 21 nM for nintedanib.” *Id.* (citing Ex. 2013, 10; Ex. 2014, 17). “Because both sunitinib and nintedanib have similar anti-VEGFR2 activity, a person of ordinary skill would have expected, on the basis of the sunitinib results described in the ‘063 provisional application, that nintedanib could effectively treat hyperemia associated with pterygium.” *Id.* at 16.

Patent Owner contends that for “an ophthalmic formulator, the most important physiochemical property is solubility.” PO Resp. 17. Patent

Owner contends that based on the available data in the literature, “the estimated water solubilities for sunitinib and nintedanib at neutral pH are 6.3 mg/L and 9.6 mg/L, respectively. A person of skill would view these numbers as the same.” PO Resp. 17 (Ex. 2005 ¶ 48). Because sunitinib and nintedanib also have similar LogP, cLogP, and pKa values, Patent Owner contends that “a person of ordinary skill would know that conventional formulation techniques would work for nintedanib.” *Id.* at 18 (Ex. 2005 ¶¶ 48–52; Ex. 2009, 67:9–20; Ex. 1005, Exhibit C).

Patent Owner contends that at the time of filing the '063 application there were two multikinase inhibitors frequently reported in the literature in topical ocular formulations, sunitinib and pazopanib. For the most basic calculated pKa, sunitinib has a value of 9.09 ± 0.10 while pazopanib has a value of 5.56 ± 0.10 . A person of skill would immediately note that nintedanib is squarely in the middle of those two numbers, confirming that conventional, known formulation techniques could be used for topical ocular formulations of nintedanib.

PO Resp. 19–20 (citing Ex. 1005, Exhibit D; Ex. 2005 ¶ 52).

(iii) Discussion

The written description requirement is found in section 112 of the patent statute, which provides that the patent’s specification must contain “a written description of the invention, and of the manner and process of making and using it.” 35 U.S.C. § 112(a). A specification that “reasonably conveys to those skilled in the art that the inventor had possession of the claimed subject matter as of the filing date” has adequate written description of the claimed invention. *Ariad Pharms., Inc. v. Eli Lilly & Co.*, 598 F.3d 1336, 1351 (Fed. Cir. 2010). “[T]he test requires an objective inquiry into

the four corners of the specification from the perspective of a person of ordinary skill in the art.” *Id.*

The case law makes clear that “[c]ompliance with the written description requirement is essentially a fact-based inquiry that will ‘necessarily vary depending on the nature of the invention claimed.’” *Enzo Biochem, Inc. v. Gen-Probe Inc.*, 323 F.3d 956, 963 (Fed. Cir. 2002) (quoting *Vas-Cath Inc. v. Mahurkar*, 935 F.2d 1555, 1562 (Fed. Cir. 1991)). In other words, what is required to meet the written description requirement “varies with the nature and scope of the invention at issue, and with the scientific and technologic knowledge already in existence.” *Capon v. Eshhar*, 418 F.3d 1349, 1357 (Fed. Cir. 2005).

A review of the ’063 provisional application establishes that application contains similar words and limitations as those disclosed in the ’820 patent. *Compare* Ex. 1003, 12 (claim 1: “A method for reducing hyperemia and symptoms in pterygia, stabilizing pterygium, inducing pterygium regression from visual axis/central cornea and treating pterygium recurrence following pterygiectomy, to an affected eye of a subject in need of such a treatment (1) a multikinase inhibitor; (2) an antimetabolite; and (3) a combination therapy of a multikinase inhibitor and an antimetabolite, in a proper dosage form.”) *with* Ex. 1001, 24:33–47 (claim 1: “A method for reducing hyperemia or symptoms thereof in pterygium in an affected eye of a subject in need of such treatment, without surgically excising a pterygium, comprising administering to the affected eye of the subject a therapeutically effective amount of a multikinase inhibitor.”). Because the ’063 provisional application describes “reducing hyperemia and symptoms in pterygia, stabilizing pterygium, inducing pterygium regression from visual

axis/central cornea,” we determined that the ’063 provisional application reasonably supports treating a patient population that has not had surgical intervention immediately preceding the treatment.

The ’063 provisional application also discloses a list of intracellular and/or cell surface protein kinases that may be the target of the multikinase inhibitor. Ex. 1003, 12 (claim 2). In addition, the ’063 provisional application provides a list 40 multikinase inhibitors, specifically listing nintedanib. *Id.* (claim 3). The disclosure of the ’063 provisional application, therefore, provides *ipsis verbis* support for the use of the multikinase inhibitor nintedanib recited in claims 4 and 5 of the ’820 patent. *Compare* Ex. 1003, 12 (claim 3 (showing a list of inhibitors) *with* Ex. 1001, 24:38–47 (showing a list of inhibitors); *see also* Ex. 1025, 465–466 (the examiner providing a list of what words do not appear in the ’063 provisional, implying that all other words find support).

We are not persuaded by Petitioner’s contention that the ’063 provisional application provides insufficient descriptive support for nintedanib as a treatment for reducing hyperemia or symptoms thereof in pterygium. We agree with Patent Owner that the ’063 provisional application discloses a limited list of 40 multikinase inhibitors, which includes nintedanib, sunitinib, and pazopanib. *See* Ex. 1003, 10; PO Resp. 14 (“The ’063 provisional application then goes on to specifically list nintedanib as one of those multikinase inhibitors.”). Patent Owner contends, and we agree, that the treatment methods recited in claims 4 and 5 are supported by the results obtained with sunitinib. PO Resp. 14–15; Ex. 1010 ¶¶ 121–147; Ex. 1014, 521. Because sunitinib worked in an animal model for pterygia, Patent Owner contends that one of ordinary skill in the art

would understand that nintedanib would work similarly because both are multikinase inhibitors with very similar pharmacological potencies related to the VEGF receptors. PO Resp. 15; Ex. 2005 ¶ 46 (“[B]ecause both share key physiochemical properties, similar known formulation approaches would work equally well for both. The potency determines how effective a drug will work when it reaches the target tissue and the physiochemical properties dictate how the drug can be formulated.”). We credit Patent Owners expert over Petitioner’s expert on this point because Roth (discussed below) further supports this finding.

We are not persuaded by Petitioner’s contention that reciting nintedanib in the specification of the ’063 provisional application is insufficient to provide written descriptive support for the effective treatment using this multikinase inhibitor. Reply 13 (citing *Nuvo Pharms.*, 923 F.3d at 1380). Unlike the facts in *Nuvo*, the record here supports Patent Owner’s position that a person of ordinary skill in the art would have expected other kinase inhibitors to work similarly to the disclosed sunitinib.

Specifically, the prior art provides that the multikinase inhibitor – pazopanib¹² – has also been formulated for ocular application and was shown to provide significant reduction in corneal neovascularization (CNV). Ex. 1008 ¶¶ 58–62; Ex. 1009, 542. Roth further establishes that systemic therapy using nintedanib, sunitinib, or sorafenib in patients with cancer provides similar efficacy with respect to neovascularization. Ex. 1011, 1055. The evidence from Roth’s systemic application further supports Patent

¹² Pazopanib is another multikinase inhibitor found in the list of inhibitors recited in the ’063 provisional application. Ex. 1003, 9.

Owner's position that sunitinib, pazopanib, and nintedanib work similarly. Therefore, the list of inhibitors recited in the '063 provisional application is more than just a random list based on wishful thinking as suggested by Petitioner.

The facts in the present appeal can also be distinguished from the facts in *Biogen International* because the disclosure of the '063 provisional application is limited to a single disease, pterygium, and treating its symptoms by topically administering an effective amount of a multikinase inhibitor. *Biogen International, GMBH v. Mylan Pharmaceutical Inc.*, 18 F.4th 1333, 1344 (Fed. Cir. Nov. 30, 2021); Tr. 11:5–7.¹³ The '063 provisional application states the treatment is based on “the support of our own in vitro and in vivo laboratory experimental data and of animal and clinical results reported in the literature, to invent treatment options for various stages of pterygia.” Ex. 1003, 9.

The prior art before the filing date of the '063 provisional application supports topical ocular application of two multikinase inhibitors sunitinib and pazopanib. Ex. 1008 ¶ 50; Ex. 1009, 542; Ex. 1010 ¶¶ 11, 121–147; Ex. 1014, 520. The application of either sunitinib or pazopanib resulted in a reduction of corneal neovascularization. Ex. 1009, 542; Ex. 1014, 524. Roth establishes that the multikinase inhibitors axitinib, nintedanib, pazopanib, sunitinib, and sorafenib behave similarly at least in systemic application to

¹³ During oral argument, Petitioner also directed our attention to *Indivior UK Limited v. Dr. Reddy's Labs. S.A.*, 18 F.4th 1323 (Fed. Cir. Nov. 24, 2021) as supporting the position that the '063 provisional application lacks written descriptive support. Tr. 13:18–19. *Indivior* is directed to the descriptive sufficiency required for finding support for a numeric range. The present claims are not directed to a range.

reduce neovascularization. *See* Ex. 1011, Abstract, 1055, 1056. Based on these disclosures before the filing date of the '063 provisional application in conjunction with the knowledge of the ordinarily skilled artisan, we agree with Patent Owner that the specification of the '063 provisional application provides sufficient written descriptive support for the effective topical ocular administration of nintedanib.

(iv) Summary

In considering the totality of the evidence, we agree with Patent Owner that there is sufficient written descriptive support in the '063 provisional application for the application of a therapeutically effective amount nintedanib for the purpose of reducing hyperemia or symptoms thereof in a patient population with pterygium.

c) Enablement

(i) Petitioner's Contentions

Petitioner argues that claims 4 and 5 are not entitled to a June 6, 2015 priority date because the '063 provisional application does not enable their full scope. *See* Pet. 30–33; Reply 28–32. Petitioner argues that “[t]he single brief description of experimental data based on sunitinib fails to demonstrate the usefulness of [nintedanib recited in] the long list of multikinase inhibitors in the ['063] provisional.” Pet. 30 (citing *Rasmusson v. SmithKline Beecham Corp.*, 413 F.3d 1318, 1323 (Fed. Cir. 2005)). Petitioner contends that “[d]ue to the diverse array of structures, targets, and properties of the multikinase inhibitors identified in the June 6, 2015 provisional,” a person of ordinary skill in the art at the time the invention was made would not have expected that the substitution of one multikinase inhibitor for another would

treat pterygium. Pet. 31 (citing Ex. 1005 ¶¶ 57–61, 67–69, 70, 74, 75). Petitioner further directs to language in the '820 patent to support the position that multikinase inhibitors are unpredictable, “[q]uite unexpectedly, the inventors found that kinase inhibitors with substantial kinase target overlaps can still have different efficacies.” Pet. 32 (citing Ex. 1001, 18:2–57; Ex. 1005 ¶¶ 73–76; Ex. 1024, 9).

Petitioner’s expert, Dr. Malone, explains that the physical structures and properties of 24 of the listed inhibitors in the '820 patent, which are also listed in the '063 provisional application, vary greatly. Ex. 1005 ¶ 66 (citing Ex. 1016), ¶ 67. The '063 provisional application describes the use of an in vivo rabbit corneal suture model to establish that treatment with sunitinib, a potent inhibitor of multiple protein kinases VEGFR2, PDGFR β , KIT, RET, CSFIR and FLT3, via topical application three times daily showed pterygium vessel regression. Ex. 1003, 10. Dr. Malone avers that he “would not expect that each of the multikinase inhibitors identified in the '820 Patent or the June 6, 2015 Provisional would [be] capable of being formulated in the same manner [as sunitinib] nor be expected to give similar efficacy results on topical administration to the eye, or any other known methods of administration such as ocular implantation” (Ex. 1005 ¶ 71), and that the necessary experimentation to create a formulation from the other listed inhibitors would be unduly extensive (*id.* ¶ 75).

(ii) Patent Owner’s Contentions

Dr. Hohman, Patent Owner’s expert, explains that a person of ordinary skill in the art would understand from the disclosure of '063 provisional application that “because sunitinib worked in an animal model for pterygia, so too would nintedanib because both are multikinase inhibitors

with very similar pharmacological potencies related to the VEGF receptors.” PO Resp. 15 (citing Ex. 2005 ¶ 46). Patent Owner contends that all that is required of the ’063 provisional application is that nintedanib, a multikinase inhibitor, be effective on some level. PO Resp. 16.

(iii) Discussion

“Undue experimentation” is a test for enablement. *In re Wands*, 858 F.2d 731 (Fed. Cir. 1988). That some experimentation is required, even a considerable amount, is not “undue” if, e.g., it is merely routine or the specification provides a reasonable amount of guidance as to the direction in which the experimentation should proceed. *Id.* at 737. Factors to consider include “(1) the quantity of experimentation necessary, (2) the amount of direction or guidance presented, (3) the presence or absence of working examples, (4) the nature of the invention, (5) the state of the prior art, (6) the relative skill of those in the art, (7) the predictability or unpredictability of the art, and (8) the breadth of the claims.” *Id.* The above “Wands factors” are illustrative, rather than mandatory, and therefore “all of the factors need not be reviewed when determining whether a disclosure is enabling.” *Enzo Biochem, Inc. v. Calgene, Inc.*, 188 F.3d 1362, 1371 (Fed. Cir. 1999). What matters for the purpose of the enablement inquiry are the particular facts being considered. *Amgen, Inc. v. Chugai Pharm. Co.*, 927 F.2d 1200, 1213 (Fed. Cir. 1991).

(1) Nature of the invention

The invention is directed to the topical application of a multikinase inhibitor – nintedanib – in a pterygium patient population in order to reduce hyperemia or symptoms of pterygium. The prior art supports the ocular application of other known structurally diverse multikinase inhibitors, in this

case sunitinib and pazopanib, as well as the use of a monoclonal antibody bevacizumab, for reducing neovascularization when directly applied to affected tissues. *See above* II.D.1.a.i–iv; Ex. 1008; Ex. 1009, Ex. 1010; Ex. 1011; Ex. 1014.

(2) Breadth of the claims

The claims are directed to the single multikinase inhibitor – nintedanib – for the treatment of pterygium in an affected eye of a subject that did not have the tissue surgically removed. The breadth of the claims is therefore narrowly focused on a single multikinase inhibitor.

(3) Working examples

The '063 provisional application explains that there are “various mechanisms involved in forming pterygia, [and] treatment strategies should target multiple pathways including proliferation, inflammation, fibrosis and angiogenesis, that have been identified in its pathogenesis. Based upon these, we have hypothesized several strategies, *with the support of our own in vitro and in vivo laboratory experimental data and of animal and clinical results reported in the literature*, to invent treatment options for various stages of pterygia.” Ex. 1003, 9 (emphasis added).

This treatment hypothesis is supported by *some experimental data conducted in our laboratory, where sunitinib*, a potent inhibitor of multiple protein kinases of VEGFR2, PDGFR β , KIT, RET, CSF1R and FLT3, *has shown to be efficacious* after three-times daily dosing for 14 days in an in vivo rabbit corneal suture model, an animal model for pterygia, in a prototype ocular solution formulation. *Similar findings have also been reported in the literature*, where topical ocular administration of sunitinib has shown more effective than bevacizumab via a subconjunctival injection, for the inhibition of corneal neovascularization in the rabbit corneal suture model.

Ex. 1003, 9 (emphasis added). Thus, the '063 provisional application provides a single working example using sunitinib in a rabbit corneal suture model and recognizes that their findings are similar to those reported in the prior art. The '063 provisional application thereby suggests that many of the details with respect to formulating these multikinase inhibitors “can be readily determined by the ordinarily skilled artisan, and can e.g., be obtained in animal models and in clinical studies reported in the literatures, for determining dosage, safety and efficacy according to standard methods known in the art.” *Id.* at 11. The '063 provisional application, therefore, evidences that much of the information relevant to practicing claims 4 and 5 would have been available to the skilled artisan by consulting the literature or relies on the use of standard art recognized methods. *Id.*; *see also* Ex. 1032, 51:10–15 (“[W]e know the potency of all of these compounds someplace. But I would imagine they have been described and one can compare the potency and solubility of those compounds to be able to predict which of them could be effective in the model and which are unlikely to be effective in the model.”).

(4) Direction or Guidance

The '063 provisional application refers to treatment of pterygium with nintedanib, where nintedanib is one inhibitor among a list of 40 inhibitors identified in the application as suitable for the purpose of treating pterygium. Ex. 1003, 9. The '063 provisional application also discloses that “some experimental data conducted in our laboratory, where sunitinib, a potent inhibitor of multiple protein kinases of VEGFR2, PDGFR β , KIT, RET, CSF1R and FLT3, has shown to be efficacious after three-times daily dosing for 14 days in an in vivo rabbit corneal suture model, an animal model for

pterygia, in a prototype ocular solution formulation.” Ex. 1003, 9–10. Just because the ’063 provisional application does not provide numbers and tables associated with their observed results does not mean we should not credit their experimental observations with regard to sunitinib. *See* Ex. 1003, 9. The ’063 provisional application explains that “[s]imilar findings have also been reported in the literature, where topical ocular administration of sunitinib has shown [to be] more effective than bevacizumab via a subconjunctival injection, for the inhibition of corneal neovascularization in the rabbit corneal suture model.” *Id.* at 9.

The recognition in the ’063 provisional application that multikinase inhibitors are effective at reducing neovascularization is further supported in the art before the effective filing date of the provisional application. For example, Pérez-Santonja and Peyman both show bevacizumab (a monoclonal antibody kinase inhibitor) and sunitinib (a small molecule multikinase inhibitor) as inhibiting corneal neovascularization after topical application. Ex. 1014, Abstract; Ex. 1010 ¶¶ 11, 121–147. King shows that topically administered pazopanib (another known multikinase inhibitor) significantly reduced development of choroidal neovascularization lesions. *See* Ex. 1008 ¶¶ 58–62. Amparo further supports this observation by showing ocular administration of pazopanib in human subjects. Ex. 1009. Specifically, Amparo’s “study shows that topical pazopanib [0.5%] is safe and well tolerated, and can effectively reduce established [corneal neovascularization] CNV as measured by invasion area, vessel length, and neovascular area.” Ex. 1009, 542. Based on these disclosures in the prior art and the disclosure in the ’063 provisional application, we determine that there is sufficient showing that the ocular application of a known

multikinase inhibitor reasonably results in a reduction of neovascularization when applied to the eye. The record supports the conclusion that structurally diverse multikinase inhibitors, in this case sunitinib and pazopanib, as well as the antibody kinase inhibitor bevacizumab function similarly in reducing neovascularization. *See above* II.D.1.a.i–iv; Ex. 1008; Ex. 1009, Ex. 1010; Ex. 1014.

Dr. Hohman, Patent Owner’s expert, explains that a person of ordinary skill in the art would understand from the disclosure of ’063 provisional application that “because sunitinib worked in an animal model for pterygia, so too would nintedanib because both are multikinase inhibitors with very similar pharmacological potencies related to the VEGF receptors.” PO Resp. 15 (citing Ex. 2005 ¶ 46). Dr. Hohman avers that the potency of sunitinib and nintedanib against VEGFR2 receptor is similar, having an IC₅₀ of 10 nM for sunitinib and 21 nM for nintedanib. Ex. 2005 ¶ 47 (citing Ex. 2013, 10; Ex. 2014, 17); *see* Ex. 1005, Exhibit B (showing IC₅₀ (nmol/L) for nintedanib is 21±13 and the K_d of sunitinib is 1.5 and pazopanib is 14). We find Dr. Hohman’s testimony to be credible because it is consistent with the various references noted above.

We disagree with Petitioner’s contention that *Rasmusson* supports its position that the ’063 provisional application does not provide an enabling disclosure for claims 4 and 5. Unlike the record in *Rasmusson*, the record here supports the position that one of skill in the art would accept that nintedanib, a multikinase inhibitor, behaves similarly to sunitinib, given the teachings in Roth’s application of these inhibitors in a systemic cancer model. *Rasmusson*, 413 F.3d at 1323–24; Ex. 1011. Roth describes that nintedanib and sunitinib have comparable efficacy in systemic application.

Ex. 1011, 1055 (“Nintedanib¹⁴ 200 mg b.i.d. monotherapy demonstrated comparable efficacy versus standard of care sunitinib as first line systemic therapy in a randomized (2:1) phase II study in patients with advanced [human renal cell carcinoma] RCC.”); *see above* II.D.1.a.iv. Roth, therefore, supports Dr. Hohman’s opinion that sunitinib and nintedanib are expected to behave similarly when applied topically. PO Resp. 15 (citing Ex. 2005 ¶ 46).

Enablement is question of law based on underlying factual inquiries. *Enzo Biochem Inc. v. Calgene Inc.*, 188 F.3d 1362, 1369 (Fed. Cir. 1999). It is well settled that patent applicants “are not required to disclose every species encompassed by their claims, even in an unpredictable art.” *In re Angstadt*, 537 F.2d 498, 502-03 (CCPA 1976). “[A] specification need not disclose what is well known in the art.” *Genentech Inc. v. Novo Nordisk A/S*, 108 F.3d 1361, 1366 (Fed. Cir. 1997).

We are not persuaded by Petitioner’s contention that efficacy for treating pterygium cannot be predicted. Reply 17 (citing Ex. 1005 ¶¶ 62–71; Ex. 1033 ¶¶ 26–30; *see also id.* ¶ 29 (“Dr. Hohman was unable to identify the efficacy of those compounds. Dr. Hohman’s inability to predict efficacy for the listed compounds is consistent with [Dr. Malone’s] opinion that the efficacy of those compounds could not be predicted from the disclosure of the June 6, 2015 Provisional.”)).

Dr. Hohman explained in his deposition that identifying which of the forty compounds listed in the ’063 provisional application would be

¹⁴ Petitioner’s expert, Dr. Malone, agrees that nintedanib is a known multikinase inhibitor and is not new. Ex. 1005 ¶ 51 (citing Ex. 1011, 1; Ex. 1012; Ex. 1013).

efficacious would not require testing but would require a review of the literature with respect to each of the listed inhibitors.

Q. Okay. So sitting here today, you can't identify for me the efficacy for any of the multikinase inhibitors on this list?

A. That's true.

Q. Okay. And you would agree that you would need to perform the testing to be able to identify for me the efficacy of each of those compounds; correct?

A. No, I don't agree with that.

Q. Okay. And you're not able to identify for me the efficacy, sitting here today, of any compound?

A. No, no, not -- not -- no. I'd have to look it up. I mean --

Q. Okay.

A. -- we know the potency of all of these compounds someplace. But I would imagine they have been described and one can compare the potency and solubility of those compounds to be able to predict which of them could be effective in the model and which are unlikely to be effective in the model.

Q. Okay. Let's -- let's talk about the testing that you would need to do to be able to verify the efficacy of a particular compound.

Are you familiar with that testing?

A. Yes.

Q. Okay. So it's a rabbit corneal suture model; is that correct?

A. That's correct.

Ex. 1032, 50:14–51:23. Dr. Hohman's deposition testimony supports the position that information about the known compounds listed the '063 provisional application can reasonably be looked up in the literature.

Tr. 37:13–38:19 (citing Ex. 1032). The multikinase inhibitors listed in the

'820 patent and the '063 provisional application were well known in the art for their systemic application in the treatment of cancers. *See* Ex. 1008, Ex. 1011, Ex. 1016. We agree with Dr. Hohman's position that it is reasonable to consult well-known information readily available in the literature for direction or guidance.

(5) Quantity of experimentation

Effective amount

Petitioner contends that there is an undue amount of work required to determine if the forty inhibitors listed in the '063 provisional application are effective at reducing hyperemia or symptoms thereof in pterygium in a subject. Specifically, Petitioner contends that this would require at least 480 rabbits and this is a considerable and undue amount of work. Reply 31–32 (citing Ex. 1005 ¶ 74 (“explaining that testing of 480 rabbits plus controls ‘would have been extremely expensive and time-consuming’”); Ex. 1032, 51:16–55:7, 57:4–16; Ex. 1033 ¶¶ 42–44); Tr. 22:6–23; Ex. 2009, 70:18–22. The test for enablement is “not merely quantitative.” *PPG Indus. v. Guardian Indus. Corp.*, 75 F.3d 1558, 1564 (Fed. Cir. 1996). On the contrary, “a considerable amount of experimentation is permissible, if it is merely routine.” *Id.*; *In re Vaeck*, 947 F.2d 488, 495 (Fed. Cir. 1991) (“That *some* experimentation may be required is not fatal; the issue is whether the amount of experimentation required is ‘undue.’”).

The remaining claims are limited to a single multikinase inhibitor – nintedanib. Therefore, the issue for those claims is whether undue experimentation is required to determine the effective amount of nintedanib for reducing hyperemia or symptoms thereof. The rabbit corneal suture model as well as the mouse model are both well-known and well-established

animal models for testing neovascularization. Ex. 1001, 6:60–64, 16:30–38, 19:6–14; Ex. 1032, 53:4–14; Ex. 1014. At the time of filing the '063 provisional application, testing multikinase inhibitors for their ability to inhibit vascularization was routine. *See* Ex. 1008; Ex. 1009; Ex. 1010; Ex. 1014; Ex. 2005 ¶ 46.

Petitioner does not establish that testing nintedanib, one of the listed inhibitors the '063 provisional application, requires an undue amount of work. *See* Ex. 2009, 71:15–21; Ex. 1005 ¶ 74. Just because the testing may be extensive and even expensive, or because it requires the use of animals, does not establish that the effort required is undue.

Patent Owner's expert, Dr. Hohman, explains that the '063 provisional application shows that sunitinib, a potent inhibitor of multiple protein kinases, has been shown to be efficacious in a rabbit corneal suture model, an animal model for pterygia, using a prototype ocular solution formulation. Ex. 2005 ¶ 46. Based on the disclosure in the specification of the '063 provisional application and what was known in the art, Dr. Hohman avers that

[a] person of skill would understand from this that because sunitinib worked in an animal model for pterygia, so too would nintedanib because both are multikinase inhibitors with very similar pharmacological potencies on the VEGF receptors. Moreover, because both share key physiochemical properties, similar known formulation approaches would work equally well for both. The potency determines how effective a drug will work when it reaches the target tissue and the physiochemical properties dictate how the drug can be formulated.

Ex. 2005 ¶ 46; *but see* Ex. 1033 ¶ 23 (Dr. Malone “disagree[s] with Dr. Hohman's opinion. Sunitinib and nintedanib have different and unique

inhibition profiles, which would have been readily apparent and known to a person of ordinary skill in the art.”).

Even though nintedanib may have a different kinase inhibition profile with respect to the other multikinase inhibitors listed in the '063 provisional application, there is still significant overlap between the kinase targets for sunitinib, pazopanib, sorafenib, and nintedanib. *See* Ex. 1001, 9:1–44 (Table 1); Ex. 1033 ¶ 24. Dr. Hohman avers that “because sunitinib worked in an animal model for pterygia, so too would nintedanib because both are multikinase inhibitors.” Ex. 2005 ¶ 46. We credit Dr. Hohman’s testimony because it finds support in the prior art.

For example, Amparo exemplified testing the multikinase inhibitor pazopanib in the form of eye drops in human subjects and shows that it is effective at reducing corneal neovascularization (CNV). Ex. 1009, 542 (“[T]opical pazopanib is safe and well tolerated, and can effectively reduce established CNV as measured by invasion area, vessel length, and neovascular area.”). Pazopanib is a multikinase inhibitor that has significant overlap with the kinase targets of not only sunitinib but also nintedanib. *See* Ex. 1001, 9:1–45 (Table 1); Ex. 1016, 18, 24, 26; Ex. 1033 ¶ 24. Peyman and Pérez-Santonja exemplify testing sunitinib topically as an eye drop for treating corneal neovascularization. *See* Ex. 1010; Ex. 1014; Ex. 2009, 57:14–58:4 (acknowledging that topical application of sunitinib and pazopanib was known). Here, the prior art shows that two different multikinase inhibitors pazopanib and sunitinib are both effective for treating corneal neovascularization. This teaching, in conjunction with the teaching that in systemic therapy applications nintedanib functioned similarly to sunitinib

and sorafenib, supports Dr. Hohman's conclusion that nintedanib is expected to function similarly when topically applied. *See* Ex. 1011, 1055.

Formulation

The next issue is whether undue experimentation is required for formulating nintedanib for topical application.

Petitioner contends that “[t]he single brief description of experimental data based on sunitinib fails to demonstrate the usefulness of the long list of multikinase inhibitors in the provisional.” Pet. 30 (citing *Rasmusson*, 413 F.3d at 1323). According to Petitioner, a number of the multikinase inhibitors listed in the '063 provisional application are barely soluble in aqueous solutions at neutral pH. Pet. 31 (citing Ex. 1005 ¶¶ 67–69; Ex. 1026, 38 (“BIBF 1120 ethanesulfonate is classified as a drug substance with low solubility according to BCS (Biopharmaceutical Classification System).”). Petitioner's expert, Dr. Malone, identified that sunitinib has a most basic pKa of 9.09, nintedanib has a most basic calculated pKa of 7.43, and sorafenib has a most basic pKa of 2.66. Ex. 1005 ¶ 67. “Based on the relationship between the pKa of a compound and the pH of a solution containing the compound, the extent to which a compound will be protonated or deprotonated in the solution can be predicted.” *Id.* (n.3). Dr. Malone concludes that in terms of their physiochemical properties, these inhibitors are not equivalent. *Id.* ¶¶ 67–67. Dr. Malone finds that solubility is also an important factor to consider when formulating a compound for topical administration. *Id.* ¶ 69. According to Dr. Malone, nintedanib has a low measured solubility in aqueous solution. Ex. 1005 ¶ 69 (citing Ex. 1026, 8, 36, 38, 39).

Patent Owner's expert, Dr. Hohman, agrees that

[a]mong the physiochemical properties, the most important physiochemical property to an ophthalmic formulator is solubility. . . . The experimentally determined water solubility at neutral pH is unknown for either sunitinib or nintedanib. But the estimated water solubilities at neutral pH can be used to guide formulation development. . . . [T]he estimated water solubilities for sunitinib and nintedanib at neutral pH are 6.3 mg/L and 9.6 mg/L, respectively. A person of skill would view these numbers as the same.

Ex. 2005 ¶ 48.

According to Dr. Hohman, "the most basic calculated pKa, sunitinib has a value of 9.09 ± 0.10 and nintedanib has a value of 7.43 ± 0.10 . Sunitinib malate has a measured pKa=8.95 and it is described as poorly soluble at neutral pH. . . . Nintedanib ethanesulfonate has a measured pKa=7.91 and is also poorly soluble at neutral pH." Ex. 2005 ¶ 51; *see also id.* ¶ 52; Ex. 1005 ¶ 69, Exhibit C–D.

According to Patent Owner, "[s]unitinib and nintedanib also have similar LogP, cLogP and pKa values – all important factors to a person of skill. Notably, the cLogP values of sunitinib and nintedanib are also very similar: 2.3 and 2.4, respectively." PO Resp. 18 (citing Ex. 2005 ¶ 48; Ex. 2009, 67: 9-20 (confirming the cLogP values are the same)).

"It is not disputed that solubility is an important consideration and that the acceptable neutral range for topical ocular formulations may extend as low as pH 6." Reply 25–26 (citing PO Resp. 17; Ex. 2009, 45:14–18; Ex. 1032, 71:15–17). The data shows that sunitinib malate has a solubility of 25 mg/mL at a pH up to 6.8. Reply 26 (citing Ex. 2013, 9). "By contrast, nintedanib's solubility at a pH of 6.8 is 0.011 mg/mL." Reply 26 (citing

Ex. 1026, 40, 125; *see* Ex. 2014, 56). Petitioner asserts that “sunitinib is over 2,000 times more soluble than nintedanib within the acceptable pH range.” Reply 27 (emphasis removed). Just because the solubility of sunitinib is significantly higher than for nintedanib at the same pH does not mean that the solubility of nintedanib is insufficient for therapeutic application.

Petitioner contends that Patent Owner’s expert, Dr. Hohman, was not able to predict the efficacy of any of the compounds listed in the ’063 provisional application, and therefore concludes that “Patent Owner had not performed any of the extensive work needed to determine any actual efficacy of nintedanib or other listed inhibitors.” Reply 30.

We disagree. There is no requirement that all the inhibitors listed in an application need to have accompanying test data in order to show enablement for the claimed subject matter. The evidence in the record supports the position that the forty inhibitors listed in the ’063 provisional application are not a random list of *all known* multikinase inhibitors at the time filing the provisional application. *See* Tr. 47:16–48:7. Davis, for example, listed 72 multikinase inhibitors yet only twenty-four of those inhibitors are also included in the list provided in the ’063 provisional application. *Compare* Ex. 1001, 8:47–56 and Ex. 1003, 12 (claims 1–3) with Ex. 1016, 17–27 (Supplemental Table 2) and Ex. 1008 ¶ 19; Ex. 1005 ¶ 66. King recites twenty-seven multikinase inhibitor compounds. Only twelve compounds from Kings’s list overlap with the compounds listed in the ’063 provisional application. *Compare* Ex. 1008 ¶ 19 *with* Ex. 1001, 8:47–56, claim 2.

We note that King recites fifteen additional multikinase inhibitors disclosed in King but not recited in the ’063 provisional application. Out of

the fifteen additional inhibitors, nine compounds have activity directed against VEGFR2. King's compounds having activity directed to VEGFR2 include: brivanib (*see* Ex. 1008 ¶ 19; Ex. 1016, 17–27 (Supplementary table 2) (reciting activity against VEGFR2)), linifanib (*see* Ex. 1008 ¶ 19; Ex. 1016, 17–27 (Supplementary table 2) (reciting activity against VEGFR2)), midostaurin (PKC412) (*see* Ex. 1008 ¶ 19), motesanib (*see* Ex. 1016, 17–27 (Supplementary table 2) (reciting activity against VEGFR2)), and OSI-930, MGCD-265, PF-337210, ENMD-2076, BMS-690514 (*see* Ex. 1008 ¶ 19). Because the list provided in the '063 provisional application omits many known multikinase inhibitors, this shows that the list is a subset of known multikinase inhibitors. *But see* Tr. 48:5–7 (Petitioner's counsel stating: "I can't identify any multiple kinase inhibitors that were known that were not included on that list. The record doesn't include any evidence as to others that may have been out there.").

According to Patent Owner, the parties' experts agree that formulating a particular multikinase inhibitor into a topical solution does not require undue experimentation:

During his deposition, Dr. Malone confirmed that it would have been obvious to a person of skill as of the filing of the '063 application to make a topical ocular formulation. Moreover, Dr. Malone agreed that formulations of multikinase inhibitors, including nintedanib, could be made by a person of skill without undue experimentation:

Q. It's your view then that Peyman contains enough information about the excipients for a topically applied ocular formulation that those formulations of the multikinase inhibitors listed in Exhibit C [of Dr. Malone's declaration] could be made without sufficient experimentation to render those formulations patentable?

MR. REED: Objection. Asked and answered.

THE WITNESS: So, I mean, there would be some experimentation required, but whether it would be undue, I don't believe so.

PO Resp. 20 (citing Ex. 2009, 64: 4–13, 49:24–50:11) (alteration in original). According to Dr. Malone, “Peyman describes a broad list of excipients and other methodology for making topical formulations. So, I mean, it would require some experimentation [to formulate multikinase inhibitors in Exhibit C], but it could be done.” Ex. 2009, 63:20–23; Ex. 1005, Exhibit C.

We agree with Patent Owner's interpretation of Dr. Malone's testimony. The fact that some experimentation is required, even a significant amount, does not convince us that an “undue” amount of effort is required to formulate nintedanib for topical application.

(iv) Summary

In considering the totality of the evidence, we agree with Patent Owner that the '063 provisional application provides an enabling disclosure for the application of a therapeutically effective amount of nintedanib for the purpose of reducing hyperemia or symptoms thereof in a patient population with pterygium, as recited in claims 4 and 5.

d) Prior determination of the earliest priority date of the '820 patent

Petitioner contends that the Office previously determined that “Patent Owner's June 6, 2015 priority document does not provide sufficient written description or enablement for the subject matter in its later-published nonprovisional applications.” Pet. 33. Petitioner directs our attention to the

prosecution history of U.S. Patent No. 10,154,994 (“the ’994 patent”), which issued from U.S. Patent Application No. 15/700,735 (“the ’735 application”) assigned to Petitioner Allgenesis. *Id.* (citing Ex. 1015, 407, 619–627, 637–654, 638–654, 678–682). The ’994 patent is assigned to Petitioner Allgenesis and is not part of the ’820 patent family, which is assigned to Patent Owner Cloudbreak and is the subject of the present IPR. Petitioner Allgenesis and Patent Owner Cloudbreak are competitors, and their patents and publications are routinely cited in each other’s prosecution.

During the prosecution of the ’735 application leading to the ’994 patent, “the examiner issued a rejection based on Patent Owner’s June 6, 2015” ’063 provisional application. Pet. 33 (citing Ex. 1015, 407). After arguing that the ’063 provisional application does not provide sufficient written description and enablement and thereby does not qualify as prior art, the examiner agreed to withdraw the rejection. *Id.* Petitioner urges us to reach the same conclusion here. *Id.* at 34.

Patent Owner opposes. PO Resp. 21–23. Patent Owner contends that the issue of priority of the ’063 provisional application discussed during prosecution of Petitioner Allgenesis’s ’735 application was not decided in Petitioner Allgenesis’s favor. PO Resp. 21. Patent Owner contends that during the prosecution of the ’735 application, “[Petitioner] Allgenesis amended its independent claims to add a limitation regarding the amount of tyloxapol and a viscosity agent” and that it was this amendment, and not the issue of priority of the ’063 patent, that ultimately resulted in the issuance of Petitioner Allgenesis patent. *Id.* at 22. Patent Owner contends that

There is no teaching of tyloxapol in the ’063 provisional application (Ex. 1003). The Office never adopted [Petitioner]

Allgenesis' argument that Ni¹⁵ was not entitled to its June 6, 2015 priority date. Instead, the Office concluded that it was the amendments that rendered the claims patentable, both in the interview summary as well as the notice of allowance.

Id. at 23 (emphasis removed).

We agree with Patent Owner that [Petitioner] Allgenesis submitted claim amendments and arguments in response to an office action rejecting the claims directed to methods treating pterygium or a symptom thereof in a subject by administering nintedanib. Ex. 1015, 612–636.

The Notice of Allowance from the prosecution history leading to the issuance of the '994 patent reads as follows:

The rejection of claims 1-8 and 11-12 under 35 U.S.C. 103 as being unpatentable over Ni, US 20170172915 A1, . . . and having an effective filing date of June 6, 2015 corresponding to provisional application US 62172063 and Bottger, US 20140235678 A1 has been withdrawn in view of Applicant's arguments regarding the *claims as amended*.

Ex. 1015, 680 (emphasis added). A review of the claim amendments leading to the Notice of Allowance shows the inclusion of additional components, specifically tyloxapol and a viscosity regulator, with the nintedanib active. *See* Ex. 1015, 613. Based on these added limitations, we agree with Patent Owner that the Notice of Allowance shows that the rejection was withdrawn in light of the amendments made by Petitioner Allgenesis and not due to any determination with respect to the priority date of the '063 provisional application. PO Resp. 21–23.

We determine that the evidence of record does not support Petitioner's argument that the Office previously determined that the '063 provisional

¹⁵ This is Patent Owner Cloudbreak's '063 provisional application.

application is not entitled to its earliest effective filing date of June 6, 2015, as reflected in the discussion and disclosures made during the prosecution of the '735 application leading to the issuance of the '994 patent, an unrelated patent family to the present '820 patent family.

e) New argument

(i) Petitioner's Contentions

Petitioner's Reply introduces new arguments and new evidence to bolster their lack of enablement arguments presented in the Petition. Petitioner contends that there are barriers for delivery of nintedanib to the target tissue that would impact the efficacy of the treatment. Reply 22–32. In addition to tears that remove the drug from the surface of the eye, barriers include “[e]fflux transporters, such as P-glycoprotein (P-gp) which is expressed in cornea and conjunctiva, would also limit drug absorption into pterygium” and reaching the intracellular target. Reply 22 (Ex. 2006, 2; Ex. 1026, 5–6). Nintedanib is a small molecule drug with an ester functional group that poses a metabolic liability. Reply 23 (citing Ex. 1026, 4). It was “well-known that conjunctival and corneal tissues possessed esterase activity.” *Id.* (citing Ex. 1033 ¶ 44; Ex. 2006, 2). Because of the extensive metabolism by esterases associated with nintedanib yielding metabolites with lower activity, one of ordinary skill in the art would recognize that this is another barrier for effectively treating patients with pterygium. *Id.* at 24 (citing Ex. 1033 ¶¶ 36–38). Petitioner notes that “sunitinib is not metabolized by esterases like nintedanib, and even when metabolized, sunitinib's major metabolite is equipotent with the drug compound itself.” *Id.* (citing Ex. 1033 ¶¶ 39, 41; Ex. 2013, 6, 35).

(ii) Patent Owner's Contentions

Patent Owner contends that Petitioner in the Reply is presenting “two new theories concerning alleged difficulties related to delivering nintedanib to target tissue.” Sur-reply 11. The new theories are the “esterase theory” and the “efflux transporter theory.” Sur-reply 11–12; Tr. 41:7–10. Patent Owner contends that “[b]oth theories are improper because each represents a new rationale for unpatentability” and there is no avenue for Patent Owner “to address them through its expert, Dr. Hohman.” Sur-reply 12 (citing Trial Practice Guide (Nov. 2019), 74).

(iii) Discussion

The legal authority applicable to Patent Owner's argument is summarized below.

Our reviewing court has explained:

It is of the utmost importance that petitioners in the IPR proceedings adhere to the requirement that the initial petition identify “with particularity” the “evidence that supports the grounds for the challenge to each claim.” 35 U.S.C. § 312(a)(3).
...

Unlike district court litigation—where parties have greater freedom to revise and develop their arguments over time and in response to newly discovered material—the expedited nature of IPRs bring with it an obligation for petitioners to make their case in their petition to institute. While the Board's requirements are strict ones, they are requirements of which petitioners are aware when they seek to institute an IPR.

Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd., 821 F.3d 1359, 1369 (Fed. Cir. 2016) (holding that petitioner's reply brief and accompanying declaration exceeded the scope of the reply under 37 C.F.R. § 42.23(b)). In a subsequent case, the Federal Circuit held that a petitioner's

reply arguments “amount to an entirely new theory . . . absent from the petition” and that “[s]hifting arguments in this fashion is foreclosed by statute, our precedent, and Board guidelines.” *Wasica Fin. GmbH v. Cont’l Auto. Sys., Inc.*, 853 F.3d 1272, 1286 (Fed. Cir. 2017) (holding that the Board did not err in declining to consider petitioner’s reply argument).

Our rules require that a petition include “[a] full statement of the reasons for the relief requested” and that “[a] reply may only respond to arguments raised in the corresponding . . . patent owner response.” 37 C.F.R. §§ 42.22(a)(2), 42.23(b).

The trial practice guide elaborates on these rules, explaining that “Petitioner may not submit new evidence or argument in reply that it could have presented earlier, e.g. to make out a prima facie case of unpatentability.” Consolidated Trial Practice Guide November 2019 (“CTPG”), 73.

Applying the legal authority, we agree with Patent Owner that Petitioner’s Reply improperly introduces new arguments and new evidence to bolster their lack of enablement arguments presented in the Petition. Petitioner does not explain why they are raising the issue of metabolism¹⁶ and efflux transporters as being a barrier to drug delivery at this stage. Reply 22–23 (citing Ex. 1033 ¶¶ 36–41). Nor does Petitioner explain why it could not have introduced this evidence earlier. Under our rules, this is

¹⁶ Because the issue of drug delivery barriers was not raised in the Petition, it is not surprising that Patent Owner’s expert, Dr. Hohman, had no opinion on this topic as he was tasked to provide an opinion about formulating multikinase inhibitors for tropical application based on the pKa and solubility arguments raised in the Petition. *See* Reply 24 (citing Ex. 1032, 20:19–21:4, 58:1–8).

procedurally improper as it belatedly presents additional rationales for questioning sufficiency of the support presented in the specification of the '063 provisional application. 37 C.F.R. § 42.23(b); CTPG 73–74. Because we determine that these new arguments are belatedly presented, we do not consider the merits of the arguments.

f) Summary

We agree with Patent Owner that the '063 provisional application enables and provides written descriptive support for the multikinase inhibitor nintedanib as claimed, and therefore, claims 4 and 5 of the '820 patent are entitled to the benefit of the June 6, 2015, priority date of the '063 provisional application.¹⁷

2. Analysis

Having determined that the '063 provisional application provides sufficient written descriptive support and enablement for the treatment of hyperemia or symptoms thereof in a patient having pterygium using the multikinase inhibitor nintedanib, the earliest effective filing date for claims 4 and 5 of the '820 patent is June 6, 2015. Allgenesis PCT therefore does not qualify as prior art against these claims of the '820 patent.

¹⁷ The effective filing date is determined on a claim-by-claim basis. *See, e.g., Lucent Techs., Inc. v. Gateway, Inc.*, 543 F.3d 710, 718 (Fed. Cir. 2008) (“Patent claims are awarded priority on a claim-by-claim basis based on the disclosure in the priority applications.”).

3. Conclusion

Because Allgenesis PCT does not qualify as prior art, Petitioner has not demonstrated that claims 4 and 5 of the '820 patent are unpatentable as anticipated by Allgenesis PCT.

E. *Alleged Obviousness over Allgenesis PCT*

Having determined that claims 4 and 5 of the '820 patent are entitled to the earliest effective filing date of June 6, 2015, the filing date of the '063 provisional application, Allgenesis PCT thereby does not qualify as prior art against the '820 patent. *See above* II.D. Accordingly, Petitioner has not met the burden of establishing that the claims are unpatentable as obvious over Allgenesis PCT.

F. *Alleged Obviousness over King and Amparo*

Petitioner provides a detailed explanation to show how the combination of King and Amparo sets forth each limitation of the challenged claims 4 and 5 to a person of ordinary skill in the art. *See* Pet. 61–69. Patent Owner opposes. PO Resp. 23–36.

1. Analysis

a) *Petitioner's Contentions*

Petitioner asserts that each limitations of claims 4 and 5 is taught by the combination of King and Amparo. *See* Pet. 61–69. Petitioner asserts that King teaches treating disorders of ocular angiogenesis and neovascularization including pterygium using suitable inhibitors. *Id.* at 62 (citing Ex. 1008 ¶¶ 1, 28). Petitioner asserts that King's suitable inhibitors target “one or more of the following receptors: VEGFR1, VEGFR2,

VEGFR3, PDGFRalpha, PDGFRbeta, c-kit, and/or FGFR (i.e., a multikinase inhibitor).” *Id.* (citing Ex. 1008 ¶ 16). Petitioner asserts that King’s inhibitors include sunitinib, sorafenib, cediranib, tivozanib, axitinib, vandetinib, nilotinib, pazopanib, and BIBF-1120 (a.k.a. nintedanib) among others. *Id.* (citing Ex. 1008 ¶¶ 16, 19–20).

Petitioner asserts that based on the teachings of King, one of ordinary skill in the art before the effective filing date of the claimed invention would have understood King to teach “treatment of pterygium including its symptoms.” Pet. 63 (citing Ex. 1005 ¶¶ 151–152). Petitioner assert that one of ordinary skill in the art would have “understood that hyperemia is a symptom of pterygium that would be treated by the methods taught by King.” *Id.* at 63 (citing Ex. 1005 ¶¶ 151–152). Petitioner asserts that “[n]one of the methods disclosed in King for treating pterygium involve the administration of a multikinase inhibitor in conjunction with surgical excision.” *Id.* (citing Ex. 1005 ¶ 153 (“King does not teach that any of its methods for treating pterygium involve the administration of a multikinase inhibitor in conjunction with surgical excision.”).)

Petitioner asserts that “Amparo describes efficacy testing of topical formulations of pazopanib, e.g., as described in King, on human subjects.” Pet. 63 (citing Ex. 1005 ¶ 155). Petitioner asserts that “[p]azopanib is a multikinase inhibitor that ‘targets VEGF and PDGF receptors and inhibits angiogenesis.’” *Id.* (citing Ex. 1009, 1). Petitioner further asserts that patients selected for the study did not have recent surgery or plan on getting surgery, and one patient selected for the study even had recurring pterygium. *Id.* (citing Ex. 1009, 2, 3 (*see* table 2 - patient 15)).

Petitioner's expert, Dr. Malone, explains that "[t]he pathogenesis of pterygium involves neovascularization—the proliferation of new blood vessels," and therefore, treating hyperemia and neovascularization substantially overlap. Ex. 1005 ¶ 150. Regarding a reason to combine the references, Petitioner's expert explains that "King and Amparo are closely-related references in the same field of research. Both references describe research on the multikinase inhibitor pazopanib and its efficacy for treating disorders of ocular angiogenesis and neovascularization including pterygium." *Id.* ¶ 140. As Petitioner's expert explains, "the references address using the same compounds (multikinase inhibitors including pazopanib) for treating the same conditions (pterygium)." *Id.* Even though "King's experimental data is limited to animal studies," one of ordinary skill in the art before the effective filing date of the claimed invention would have looked to Amparo for providing "further insight in the development of a pharmaceutical therapy for pterygium and its symptoms" using multikinase inhibitors in human subjects. *Id.* ¶ 141.

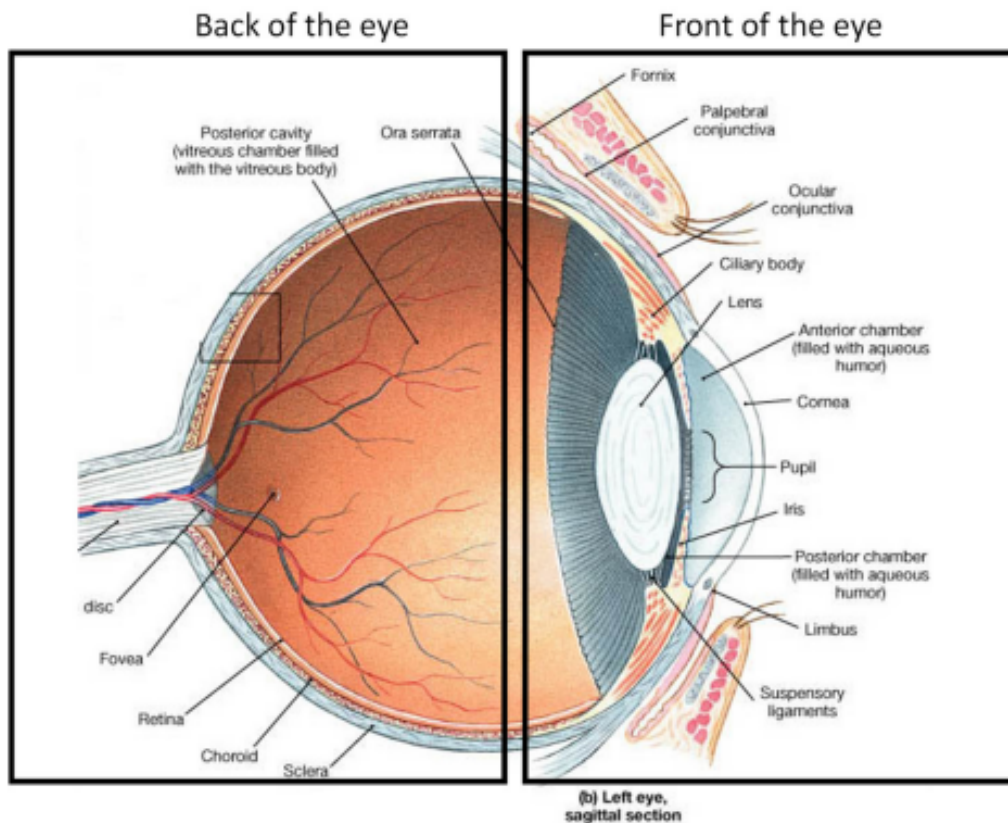
b) Patent Owner's Contentions

Patent Owner contends that the combination of King and Amparo would not render claims 4 and 5 obvious. PO Resp. 23–36. Patent Owner contends that King mischaracterizes pterygium, which is "associated with the conjunctiva and cornea, both of which are anterior ocular tissues. It is not a retinal vascular disease." PO Resp. 25. Patent Owner contends that King's experimental methods are directed at establishing whether the multikinase inhibitor pazopanib could be transported to the posterior of the eye. *Id.* at 26. According to Patent Owner, King's experimental model is not a model for

pterygium. *Id.* at 26 (Ex. 1002 ¶ 152; Ex. 2005 ¶¶ 32–33). Patent Owner directs our attention to King’s teaching that “topical administration of pazopanib to one of the eyes resulted in detectable plasma concentrations of pazopanib, as well as detectable levels of pazopanib in the sclera, choroid, and retina of both the treated and untreated (fellow) eye.” *Id.* at 28 (Ex. 1008 ¶ 62; Ex. 2005 ¶ 61). Patent Owner contends that “a person of ordinary skill would have recognized that King teaches nothing regarding whether pazopanib, applied topically, could be delivered to the conjunctiva or cornea.” *Id.* at 29.

Patent Owner further contends that one of ordinary skill in the art would not have been motivated to combine King and Amparo. “As Dr. Hohman explains, King relies on systemic absorption of pazopanib in order to deliver pazopanib to the retina and choroid, noting that it provides an ‘important contribution’ for delivery of pazopanib to retinal and choroidal tissue.” PO Resp. 30 (Ex. 2005 ¶ 64; Ex. 1008, ¶ 66).

Dr. Hohman explains that the conjunctiva and cornea are tissues located in the anterior portion of the eye, and that pterygium is considered a disease relating to anterior ocular tissues, whereas ocular tissues such as choroidal and retinal tissues are located in the posterior portion of the eye. Ex. 2005 ¶ 32. Below is a figure, reproduced from Dr. Hohman’s declaration, showing a sagittal section of the left eye identifying various structures.



The figure above shows the various anatomical features associated with the eye. Ex. 2005 ¶ 32 (citing Ex. 2006, 1; Ex. 2007, 3). Dr. Hohman’s declaration explains that “anterior ocular diseases typically are treated topically by administering eye drops.” *Id.* ¶ 33 (citing Ex. 2006, 3; Ex. 2007, 3).

Patent Owner contends that because King is directed at delivering pazopanib to the posterior portion of the eye by systemic absorption, a person of ordinary skill in the art would not have had reason to combine it with Amparo, which took specific measures to avoid systemic absorption. *See* PO Resp. 29–30. Patent Owner contends that King and Amparo are odds with each other and therefore there is no motivation to combine the references. PO Resp. 31 (citing *In re Ratti*, 270 F.2d 810, 813 (CCPA 1959))

(no motivation to combine references where proposed combination would change the basic principle under which the primary reference operated)). Patent Owner contends that “neither King nor Amparo teaches administering any agent to treat pterygia without surgical excision. Amparo, in fact, teaches the opposite.” Sur-reply 21.

c) Discussion

(i) Motivation to combine

Inquiry into motivation to combine considers whether a person having skill in the art would have believed that there was a reason for reaching the claimed invention in the first place. *Intelligent Bio-Systems* 821 F.3d at 1368; *see also KSR Int’l Co. v. Teleflex Inc.*, 550 U.S. 398, 418 (2007) (“it can be important to identify a reason that would have prompted a person of ordinary skill in the relevant field to combine the elements in the way the claimed new invention does”).

Petitioner relies on King for teaching the application of pazopanib in the form of an eye drops. Pet. 62–69. Petitioner contends that King does not mischaracterize pterygium as a posterior ocular disorder, but instead identifies pterygium as a disorder of ocular angiogenesis and lists it along with several other anterior ocular disorders that may be treated using King’s methods. Reply 34 (“[P]terygium is correctly identified [in King] as a “disorder of ocular angiogenesis” and is listed with several other anterior ocular disorders that may be treated using King’s methods.” (Ex. 1008 ¶ 28; *see also* Ex. 1032, 76:11-77:21)).

According to Petitioner, King’s teachings are not limited to systemic delivery. “King provides examples of topical ocular formulations of

multikinase inhibitors in the form of a solution. King specifically teaches topical eye drops formulated as a solution comprising pazopanib and pharmaceutically acceptable excipients,” as well as testing the topical ocular formulations. Pet. 65 (Ex. 1008 ¶¶ 48, 58–59). King is thus “not limited to oral administration.” Reply 33 (citing Ex. 1008 ¶¶ 48, 50, 58–62; Ex. 1005, ¶ 152). “Dr. Hohman[, Patent Owner’s expert,] conceded that King’s experimental results showed that drug also was delivered “to the conjunctiva...separate from the systemic delivery.” Reply 34 (citing Ex. 1032, 78:5-19).

“Petitioner’s proposed combination relies on the topical route of administration [of a multikinase inhibitor] taught by King and Amparo—not systemic absorption.” Reply 35 (citing Pet. 65; Ex. 1033 ¶ 55). We agree with Petitioner, and find there is sufficient motivation to combine King and Amparo based on each references’ teachings of topically applying the multikinase inhibitor pazopanib for the purpose of inhibiting neovascularization.

Having determined that there is motivation to combine the references, we also agree with Petitioner that selection of nintedanib from King’s disclosed list would have been obvious because King identifies the inhibitors as suitable alternatives. Pet. 65 (“It would have been obvious to a POSITA to use nintedanib, as taught by King, as the multikinase inhibitor in a topical ocular formulations as taught by King and Amparo.”); Tr. 7:7–9 (“We’re relying upon the expressed teaching in King that nintedanib was a suitable inhibitor for the treatment of pterygium.”). “[P]icking and choosing [from a list] may be entirely proper in the making of a 103, obviousness rejection, where the applicant must be afforded an opportunity to rebut with objective

evidence any inference of obviousness.” *In re Arkley*, 455 F.2d 586, 588 (CCPA 1972).

(ii) Missing limitation: “without surgical excision of the pterygium”

Patent Owner contends that claims 4 and 5 are directed at treating a patient “without surgical excision of the pterygium, which was the conventional treatment for pterygia as of June 6, 2015.” PO Resp. 34; Sur-reply 21. Patent Owner notes that Amparo studied safety not efficacy of the pazopanib administration, and explicitly cautions against drawing conclusions regarding efficacy of the topical pazopanib administration. *Id.* at 32. According to Patent Owner, Amparo provides data for only three patients, none of whom had pterygia. Based on this limited disclosure, “Dr. Hohman explains, a person of ordinary skill, reading Amparo, would not have been able to determine whether topically applied pazopanib could successfully treat pterygium.” PO Resp. 33 (citing Ex. 2005 ¶ 70).

Petitioner contends that “[n]one of the methods disclosed in King for treating pterygium involve the administration of a multikinase inhibitor in conjunction with surgical excision.” Pet. 63 (citing Ex. 1008 ¶ 153). Furthermore, Petitioner asserts that the patients included in Amparo’s treatment protocol were selected because they did not have either recent or planned surgery. *Id.*; *see also* Reply 36 (“Nothing in Amparo disparages treating patient without recent or planned surgery” (citing Ex. 1033 ¶ 57)).

Amparo selected “[o]nly patients with clinically stable CNV [corneal neovascularization] that extended at least 1-mm beyond the limbus . . . [and excluded patients that had] ocular surgery on the studied eye within the previous 3 months.” Ex. 1009, 2. This means that Patient-15 in Amparo, the

one with recurring pterygium, did not have surgery at least within three months prior to participation in the trial. *Id.* at 3 (Table 2); Ex. 1005 ¶ 155. This reasonably supports the position that Amparo does not surgically excise¹⁸ the pterygium while at the same time applying the pazopanib eye drop therapy. While Amparo may have provided before and after pictures for only three patients (*see* Ex. 1009, 5–6), Amparo concludes that “[t]reatment with topical pazopanib led to a 22% decrease of corneal neovessels (NA) and NA is a function of vessel caliber and length.” Ex. 1009, 6, 7 (“[T]he results of this study indicate that topical treatment with pazopanib 0.5% leads to a statistically significant reduction in CNV based on decrease in NA, VL, and IA.”). Treatment as disclosed in the ’820 patent encompasses prevention, arresting development, and regression. *See* Ex. 1001, 8:16–27. A reduction of corneal vessel with treatment in the patient population as described in Amparo, which includes a patient having recurring pterygium, reasonably supports reducing hyperemia or symptoms of pterygium as encompassed by claims 4 and 5 of the ’820 patent. *See* Ex. 1005 ¶ 155.

Accordingly, we conclude that Petitioner has demonstrated that the limitation “without surgical excision of the pterygium” is sufficiently supported in the teachings of King and Amparo.

¹⁸ The limitation “without surgically excising a pterygium” defines a particular patient population (*see above* II.B.2). The limitation, however, does not describe a patient population that has never had surgery; it only describes a population that currently has a pterygium that is not excised prior to, or in connection with, multikinase inhibitor treatment.

(iii) Expectation of success

King teaches applying “therapeutically effective amounts of the suitable inhibitor, and one or more pharmaceutically acceptable carriers, diluents, or excipients” to the eye of an animal subject. Ex. 1008 ¶ 30. King teaches “[t]he suitable inhibitor can be various inhibitors that inhibit one or more of the following receptors: VEGFR1, VEGFR2, VEGFR3, PDGFRalpha, PDGFRbeta, c-kit, and/or FGFR.” Ex. 1008 ¶ 19. King includes nintedanib (BIBF-1120) in a list of suitable inhibitors. Ex. 1008 ¶ 19.

King teaches the “[t]opical [application of] eye drops [that] were formulated in a buffered 7% cyclodextrin solution containing 5 mg/ml pazopanib.” Ex. 1008 ¶ 48. “Pazopanib (approximately 30 µl of a 5 mg/ml sterile-filtered solution) was administered twice daily [onto the eyes of rats]. Animals of the control group received a vehicle only.” Ex. 1008 ¶ 50. King teaches that in their choroidal neovascularization rat model, “[w]hen areas of vessel leakage were followed up by fluorescence angiography from post laser days 7 to 14, topically administered pazopanib significantly reduced development of [choroidal neovascularization] lesions. In contrast, leakage of [choroidal neovascularization] lesions continued to progress in eyes of the control group treated with the vehicle.” Ex. 1008 ¶ 59, *see also id.* ¶¶ 60–61. King, therefore, teaches that topical administration of pazopanib resulted in a reduction in the lesion size in treated eyes.

Amparo teaches using “[o]nly patients with clinically stable CNV that extended at least 1-mm beyond the limbus.” *Id.* at 2. Patients were instructed to apply one drop of pazopanib 0.5% solution into the eye four times per day for 3 weeks. *Id.* The patient population tested included a patient having a

diagnosis of “MGD, rosacea, DES, [and] recurrent pterygium.” *Id.* at 3 (table 2, patient 15). “The primary efficacy outcome measures were the size and extent of CNV. The efficacy of topical pazopanib treatment on CNV was evaluated by comparing corneal photographs acquired at baseline with photographs acquired on the follow-up visits.” *Id.* at 2. “Treatment with topical pazopanib led to a 22% decrease of corneal neovessels (NA) and NA is a function of vessel caliber and length.” *Id.* at 6. The study shows that “topical pazopanib is safe and well tolerated, and can effectively reduce established CNV as measured by invasion area, vessel length, and neovascular area.” *Id.*

“The reasonable expectation of success requirement refers to the likelihood of success in combining references *to meet the limitations of the claimed invention.*” *Intelligent Bio-Systems, Inc. v. Illumina Cambridge Ltd.*, 821 F.3d 1359, 1367 (Fed. Cir. 2016) (emphasis added). Expectation of success is assessed from the perspective of a person of ordinary skill in the art, at the time the invention was made. *See Life Techs. Inc. v. Clontech Labs. Inc.*, 224 F.3d 1320, 1326 (Fed. Cir. 2000).

It is obvious to those skilled in the art to substitute one known equivalent for another. *See In re Omeprazole Patent Litig.*, 483 F.3d 1364, 1374 (Fed. Cir. 2007) (“[T]his court finds no . . . error in [the] conclusion that it would have been obvious to one skilled in the art to substitute one ARC [alkaline reactive compound] for another.”). “Conclusive proof of efficacy is not necessary to show obviousness. All that is required is a reasonable expectation of success.” *Hoffman-La Roche Inc. v. Apotex Inc.*, 748 F.3d 1326, 1331 (Fed. Cir. 2014).

Based on the teachings of King and Amparo, we agree with Petitioner that it would have been obvious to select any one of the multikinase inhibitors listed in King, including nintedanib, and apply them to the animal model disclosed in King or the human subjects as taught in Amparo. Pet. 65 (King “teaches that BIBF-1120 is a suitable inhibitor. . . . BIBF-1120 is another name for nintedanib. . . . It would have been obvious to a [person of ordinary skill in the art] to use nintedanib, as taught by King, as the multikinase inhibitor in a topical ocular formulations as taught by King and Amparo.” (citing Ex. 1008 ¶ 16; Ex. 1011, 1; Ex. 1005 ¶¶ 159–163)). Petitioner emphasizes that its position for the selection is not based on the inhibitors being suitable equivalents but instead relies on King’s express disclosure that any one of the inhibitors can be applied. Pet. 65 (“It would have been obvious to a POSITA to use nintedanib, as taught by King, as the multikinase inhibitor in a topical ocular formulations as taught by King and Amparo.”); Tr. 7:7–9 (“We’re relying upon the expressed teaching in King that nintedanib was a suitable inhibitor for the treatment of pterygium.”).

Patent Owner argues that King is directed to oral administration and treating diseases associated with the posterior of the eye, “specifically, retinal and choroidal ocular tissue.” PO Resp. 24 (citing Ex. 2005, ¶ 56; Ex. 2006, 1; Ex. 2007, 3). Patent Owner argues that King’s experiments are directed as neovascularization after rupture of Bruch’s membrane,¹⁹ and “a

¹⁹ King’s research using pazopanib is directed to eye disorders in which the neovascularization plays a role in age-related macular degeneration and the vision loss resulting from choroidal neovascularization (CNV). Ex. 1008 ¶ 3, code (57). Amparo on the other hand applies pazopanib in the treatment of corneal neovascularization (CNV). Ex. 1009, 1. We note that CNV has different meaning in the two references. CNV in King is referencing the

useful model for neovascular age-related macular degeneration.” *Id.* at 26. King’s experimental model is not, however, a model for pterygium. *Id.* According to Patent Owner, King studied systemic concertation of pazopanib after topical administration. King’s “topical administration of pazopanib to one of the eyes resulted in detectable plasma concentrations of pazopanib, as well as detectable levels of pazopanib in the sclera, choroid, and retina of both the treated and untreated (fellow) eye.” *Id.* at 28. According to Patent Owner, “King teaches nothing regarding whether pazopanib, applied topically, could be delivered to the conjunctiva or cornea.” *Id.* at 29.

Patent Owner fails to address King’s experimental treatment using a well-known animal model. Ex. 1005 ¶ 152. King may have been focused on showing the feasibility for oral administration of multikinase inhibitors for the purpose of treating angiogenesis in the eye, but the reference is art for all that it discloses and suggests to one of ordinary skill in the art. *In re Lamberti*, 545 F.2d 747, 750 (CCPA 1976); *see also In re Susi*, 440 F.2d 442, 446 n.3 (CCPA 1971) (disclosed examples and preferred embodiments do not constitute a teaching away from a broader disclosure or non-preferred embodiments.).

King’s experimental treatment includes administering pazopanib solution (approximately 30 µl of a 5 mg/ml sterile-filtered solution) twice

back of the eye, while CNV in Amparo is referencing the front of the eye. *See* Ex. 2005 ¶ 32 (showing a figure depicting the elements of the front and back structure of the eye). Amparo recognizes that “pazopanib has been shown to inhibit laser-induced choroidal NV” in an experimental model such as the one presented in King. Ex. 1009, 1.

daily to the eye of the animals. *See* Ex. 1008 ¶¶ 48–50, 58–62. Based on this disclosure, we are not persuaded that King is limited to oral administration. In one test, King topically instilled the pazopanib solution onto the eye of an animal, measured the plasma levels as well as levels in the treated eye cup tissues (sclera, choroid, and retina), and noted that that pazopanib was also detected in the eye that did not receive eye drops (the fellow (OD) eye). Ex. 1008 ¶ 62. We find no evidence in the record from which to conclude that topically applying the pazopanib solution to the eye, as disclosed in King, would not make the solution available to the tissues that are associated with the front (i.e., anterior) of the eye, which includes the cornea as well as the ocular conjunctiva. *See e.g.* Ex. 2005 ¶ 32 (providing a figure detailing the structural components of the eye); Reply 34 (“On cross-examination, Dr. Hohman conceded that King’s experimental results showed that drug also was delivered ‘to the conjunctiva...separate from the systemic delivery.’ Ex. 1032, 78:5-19 (admitting that in King this “topical ocular formulation diffuses . . . to the conjunctiva and eventually into the retina. So there’s a diffusion component, which is separate from the systemic delivery.”).”). Accordingly, we are not persuaded by Patent Owner’s contention that King is limited to systemic absorption.

Furthermore, Petitioner does not rely on the oral administration route taught in King to establish a reason to combine the references. Instead Petitioner is relying on the experimental animal model data taught in King as a reason to combine King with Amparo. *See* Pet. 63 (citing Ex. 1005 ¶¶ 152–152). Amparo acknowledged that “pazopanib [formulated as an eye drop] has been shown to inhibit laser-induced choroidal NV” (neovascularization), the experimental model studied in King. Ex. 1009, 1.

To specifically study the effects of topical pazopanib on the corneal tissue, i.e., tissue associated with the front of the eye, Amparo plugged the superior and inferior puncta of the study eye to avoid any systemic absorption. Ex. 1009, 8. Amparo concluded that “[t]reatment with topical pazopanib led to a 22% decrease of corneal neovessels (NA) and NA is a function of vessel caliber and length.” *Id.* There is no reason to conclude that the other inhibitors listed in King would not function similarly at inhibiting neovascularization.

Accordingly, we conclude that Petitioner has demonstrated that the that based on the combination of King and Amparo there is reasonable expectation of success that nintedanib would result a reduction in neovessels when topically applied.

(iv) Unexpected results

We next determine if there are any secondary considerations that when considered against the totality of the evidence that leads us to conclude that the claims are patentable. Sur-reply 5.

Petitioner directs our attention to the “unexpected nature” of nintedanib in the context of their enablement arguments to establish that multikinase inhibitors are not equivalent. Pet. 31–32. Petitioner contends that Patent Owner “admitted that sunitinib[, pazopanib,] and nintedanib are not equivalent or substitutable during prosecution of a continuation of the ’820 Patent (Appl. No. 16/210,896),” because “nintedanib showed unexpected results as compared with sunitinib—the only compound tested at the time the ’063 provisional was filed.” Reply 19 (citing Ex. 1023, 9–11) (emphasis removed). Petitioner having directed our attention to the unexpected properties of nintedanib at least in the context of their

enablement discussion, we now cannot ignore this evidence in the context of an obviousness determination.

The panel can draw its own inferences from the arguments and evidence before it. *See Rovalma, S.A. v. Bohler-Edelstahl GmbH & Co. KG*, 856 F.3d 1019, 1027 (Fed. Cir. 2017) (noting that the Board is not precluded “from relying on arguments made by a party and doing its job, as adjudicator, of drawing its own inferences and conclusions from those arguments”). “After a prima facie case of obviousness has been made and rebuttal evidence submitted, all the evidence must be considered anew.” *In re Eli Lilly & Co.*, 902 F.2d 943, 945 (Fed. Cir. 1990) (citing *In re Piasecki*, 745 F.2d 1468, 1472 (Fed. Cir. 1984)).

According to the '820 patent:

Nintedanib and sunitinib both inhibit the key VEGFR and PDGFR families with a few non-overlapping targets (see Table 1). But the unique set of kinase targets of nintedanib seems to make it more effective and safe than sunitinib, indicating that nintedanib may be one of the most powerful multikinase inhibitors for reducing corneal neovascularization. The unique profile for the inhibition of the following intracellular and/or cell surface protein kinases at the maximum in vitro IC₅₀ are specified as VEGFR (1, 2, 3) (IC₅₀<50 nM), PDGFR (α,β) (IC₅₀<100 nM), FGFR (1, 2, 3) (IC₅₀<150 nM), FGFR4 (IC₅₀<1000 nM), FLT3 (IC₅₀<50 nM), Lek (IC₅₀<50 nM), Lyn (IC₅₀<200 nM), and Src (IC₅₀<200 nM). *Three targets of FGFR4, Lyn and Src, are not inhibited by sunitinib or other common kinases inhibitors and these may differentiate nintedanib from sunitinib and others.* In addition, the nintedanib's potencies on FGFR1-3 are substantially higher than that of sunitinib that may also contribute to the *superior efficacy of nintedanib in the cornea suture rabbit model.*

Ex. 1001, 12:1–20 (emphasis added).

Quite unexpectedly, the inventors found that kinase inhibitors with substantial kinase target overlaps can still have different efficacies. Although nintedanib and sunitinib both inhibit the key VEGFR family, they have several non-overlapping targets. As shown in Table 1, the target profile of nintedanib is different from that of sunitinib as well as several other MKI's considered very similar in the research community. The special set of kinase targets of nintedanib seems offer at least two surprising advantages: 1) it makes nintedanib a highly effective inhibitor of neovascularization, more so than sunitinib, as demonstrated in the rabbit suture model; 2) it targets fewer kinases than sunitinib, allowing for better safety margins and higher doses. Accordingly, it is understood that *nintedanib has demonstrated an improved efficacy and a better safety profile as compared with sunitinib.*

Ex. 1001, 18:2–18 (emphasis added).

We understand that the '063 provisional application does not provide any data with respect to nintedanib in the corneal suture model, nor does the '063 provisional application recognize that nintedanib shows surprising efficacy. *Compare* Ex. 1003 *with* Ex. 1001, 11: 26–27 (“Interestingly, nintedanib showed a clear trend of superiority to sunitinib in reducing neovascular area.”). Not every property of a compound, however, needs to be fully recognized as of the filing date of the patent application. *Genetics Institute, LLC v. Novartis Vaccines and Diagnostics, Inc.*, 655 F.3d 1291, 1307 (2011) (citing *Knoll Pharm. Co. v. Teva Pharms. USA, Inc.*, 367 F.3d 1381, 1385 (Fed. Cir. 2004) (“There is no requirement that an invention’s properties and advantages were fully known before the patent application was filed, or that the patent application contains all of the work done in studying the invention, in order for that work to be introduced into evidence in response to litigation attack.”)). Indeed, our reviewing court has held “that

evidence of unexpected results may be used to rebut a case of *prima facie* obviousness even if that evidence was obtained after the patent’s filing or issue date.” *Id.*

As discussed above, one of skill in the art would expect multikinase inhibitors to work similarly. *See above* F.1.c.iii. The ’820 patent, however, provides evidence of unexpected results by comparing nintedanib to the closest prior art sunitinib.²⁰ Specifically, the ’820 patent discloses that nintedanib provides “improved efficacy and better safety profile as compared to sunitinib.” Ex. 1001, 18:17–18. The ’820 patent shows that in a rabbit cornea suture model the neovascular area is substantially reduced in the eyes treated with nintedanib as compared to sunitinib or the vehicle control. Ex. 1001, 11:17–24. Table 2 showing these results is reproduced below.

TABLE 2

	Neovascular Area Mean Percent Change over Day 7 Baseline (– sign indicates decrease)					
	0.05% sunitinib TID	0.2% nintedanib BID	0.2% nintedanib TID	0.05% nintedanib BID	0.05% nintedanib TID	Vehicle TID
Day 10 (after 2 days of treatment)	10.80	12.09	–16.84	64.08	1.99	199.68
P value vs sunitinib		0.96	0.247	0.090	0.783	0.003
Day 12 (after 4 days of treatment)	49.72	37.15	–13.80	45.56	–27.87	189.01
P value vs sunitinib		0.67	0.074	0.917	0.016	0.079
Day 14 (after 6 days of treatment)	98.38	73.51	–1.51	99.48	32.66	349.43
P value vs sunitinib		0.54	0.009	0.984	0.093	0.057

²⁰ King lists sunitinib as an art recognized suitable inhibitor. Ex. 1008 ¶ 19; *see also* Ex. 1011, 1056 (Inhibitors such as sunitinib, sorafenib, pazopanib or axitinib have been integrated into standard clinical practice)

Ex. 1001, 11. Table 2 shows three observation days comparing 0.05% sunitinib TID with 0.05% nintedanib. On all three days, 0.05% nintedanib TID shows significant improvements over 0.05% sunitinib TID. For example, on day 12, the 0.05% nintedanib TID group had a negative value (indicating a reduction in neovascular area), while the 0.05% sunitinib TID group had a positive number (indicating a growth in neovascular area) and the difference between the two is also statistically significant with a P value of 0.016. The comparative data of measuring the reduction in neovascularization in an established animal model as shown the '820 patent provides a sufficient nexus to the claimed method of reducing hyperemia. The data presented in Table 2 (shown above) shows that nintedanib functions unexpectedly better than sunitinib.

The '820 patent further explains that nintedanib inhibits three kinase targets, FGFR4, Lyn and Src, that are not inhibited by sunitinib, pazopanib, or other common kinases inhibitors. Ex. 1001, 12:14–17; *see also id.* at 11 (Table 1). In addition, “nintedanib’s potencies on FGFR1-3 are substantially higher than that of sunitinib that may also contribute to the superior efficacy of nintedanib in the cornea suture rabbit model.” *Id.* 12:17–20. Given these surprising and unexpected results that nintedanib has improved efficacy and a better safety profile compared to sunitinib as reported in the '820 patent specification convinces us that on balance the weight of the evidence supports a finding that claims 4 and 5 have not been shown to be unpatentable.

After considering all the evidence anew before us, we conclude that Petitioner has not provided sufficient evidence to demonstrate that claims 4 and 5 are unpatentable.

2. Conclusion

Upon weighing all the evidence, including the competing testimony, the reasoning to combine the references, and the evidence of secondary considerations in the form of surprising unexpected results, we determine that Petitioner has not shown that claims 4 and 5 of the '820 patent are unpatentable.

III. CONCLUSION

For the foregoing reasons, we conclude that Petitioner has not established by a preponderance of the evidence that claims 4 and 5 of the '820 patent are unpatentable.

Claims	35 U.S.C. §	Reference(s)/Basis	Claims Shown Unpatentable	Claims Not Shown Unpatentable
4, 5	102	Allgenesis PCT		4, 5
4, 5	103	Allgenesis PCT		4, 5
4, 5	103	King, Amparo		4, 5

IV. ORDER

Accordingly, it is:

ORDERED that claims 4 and 5 of the '820 patent have not been shown to be unpatentable; and

FURTHER ORDERED that any party seeking judicial review must comply with the notice and service requirements of 37 C.F.R. § 90.2.

IPR2020-01438
Patent 10,149,820 B2

PETITIONER

David A. Reed
Kristopher L. Reed
Kilpatrick Townsend & Stockton LLP
dreed@kilpatricktownsend.com
kreed@kilpatricktownsend.com

PATENT OWNER

Dorothy P. Whelan
Chad Shear
Fish & Richardson P.C.
3200 RBC Plaza
whelan@fr.com
shear@fr.com